

Chapter - 1**Multiple Choice Questions***(Encircle the correct answer choice)*

- For any complex number z , it is always true that $|z|$ is equal to
 (a) $|\bar{z}|$ (b) $|-z|$ (c) $|\bar{-z}|$ (d) all of these
- If z_1 and z_2 are any two complex number, then
 (a) $|z_1 + z_2| < |z_1| + |z_2|$ (b) $|z_1 + z_2| \leq |z_1| + |z_2|$
 (c) $|z_1 - z_2| < |z_1| - |z_2|$ (d) $|z_1 + z_2| \geq |z_1| + |z_2|$
- If z_1 and z_2 are two complex number then
 (a) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ (b) $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$
 (c) $|z_1 z_2| = |z_1| |z_2|$ (d) all of these
- The numbers which can be put in the form of $\frac{p}{q}$ $p, q \in \mathbb{Z}, q \neq 0$ are
 (a) Rational numbers (b) Irrational numbers
 (c) Natural numbers (d) Integers
- The numbers which cannot be written in the form of $\frac{p}{q}$ $p, q \in \mathbb{Z}, q \neq 0$ are
 (a) Rational numbers (b) Irrational numbers
 (c) Complex numbers (d) Whole numbers
- A decimal which has only a finite numbers of digits in its decimal part is called.
 (a) Terminating decimal (b) Non-terminating decimal
 (c) Recurring decimal (d) Non recurring
- A decimal in which one or more digits repeat indefinitely in its decimal part is called
 (a) Terminating decimal (b) Periodic decimal
 (c) Infinite set (d) Repeated number
- Every recurring decimal is
 (a) a rational number (b) an Irrational number
 (c) a prime integer (d) a whole number
- A non terminating and a non recurring decimal is
 (a) a rational number (b) an Irrational number
 (c) Periodic number (d) a sequence
- 5.333.....is
 (a) Rational (b) Irrational (c) an Integer (d) a prime integer
- π is
 (a) Rational (b) Irrational (c) Natural number (d) None
- $\frac{22}{7}$ is
 (a) Rational (b) Irrational (c) an Integer (d) a whole number

13. π is the ratio
 - (a) $\frac{\text{circumference of circle}}{\text{length of diameter}}$
 - (b) $\frac{\text{circumference of circle}}{\text{length of Radius}}$
 - (c) $\frac{\text{length of diameter}}{\text{circumference of circle}}$
 - (d) $\frac{\text{length of Radius}}{\text{circumference of circle}}$
14. Every Integer is also a
 - (a) a rational number
 - (b) an Irrational number
 - (c) a Natural number
 - (d) a decimal number
15. If n is a prime number, then $\sqrt{n}$ is
 - (a) a rational number
 - (b) an Irrational number
 - (c) an Integer
 - (d) periodic number
16. If n is a negative number, then $\sqrt{n}$ is
 - (a) a rational number
 - (b) an Irrational number
 - (c) only negative integer
 - (d) a pure Imaginary
17. The number '0' is
 - (a) a rational number
 - (b) an integer
 - (c) Even number
 - (d) all of these
18. The number '0' is
 - (a) a non positive integer
 - (b) a non negative integer
 - (c) Real number
 - (d) whole number
 - (e) all of these
19. If $a, b \in R$ and $(a + b) \in R$ then this property of real numbers is
 - (a) Closure property w. r. t. +
 - (b) Commutative property w. r. t. +
 - (c) Associative property w. r. t. +
 - (d) Additive property
20. For $a, b \in R$ if $a + b = b + a$, then this property is called
 - (a) Closure property w. r. t. +
 - (b) commutative property w. r. t. +
 - (c) Associative property w. r. t. +
 - (d) Distributive property
21. Multiplicative Inverse of 0 is
 - (a) 0
 - (b) Any real number
 - (c) Not defined
 - (d) 1
22. If a is any non-zero real number, then its multiplicative inverse is
 - (a) $-a$
 - (b) $\frac{1}{a}$
 - (c) $-\frac{1}{a}$
 - (d) Not defined
23. For all $a \in R$, $a = a$ is property
 - (a) Reflexive
 - (b) Symmetric
 - (c) Transitive
 - (d) Trichotomy
24. For all $a, b \in R$, $a = b \Rightarrow b = a$ is called Property
 - (a) Reflexive
 - (b) Transitive
 - (c) symmetric
 - (d) Trichotomy
25. For $a, b, c \in R$ if $a = b, b = c \Rightarrow a = c$, then it is property
 - (a) Transitive
 - (b) Trichotomy
 - (c) cancellation
 - (d) symmetric
26. For $a, b, c \in R$ $a = b \Rightarrow a + c = b + c$, then it is property
 - (a) Transitive
 - (b) Trichotomy
 - (c) cancellation
 - (d) Additive
27. For $a, b, c \in R$ $a + c = b + c \Rightarrow a = b$, then it is Property
 - (a) Transitive
 - (b) Trichotomy
 - (c) cancellation
 - (d) Additive

28. For $a, b, c \in R$ $a = b \Rightarrow ac = bc$, then it is property
 (a) Commutative (b) Closure (c) Transitive (d) Multiplicative
29. For $a, b, c \in R$ and $a > b, b > c \Rightarrow a > c$, then it is property
 (a) Transitive (b) Trichotomy (c) Cancellation (d) Inverse
30. For $a, b, c \in R$, if $a < b$ and $c > 0$, then which is true
 (a) $a + c > b + c$ (b) $ac > bc$ (c) $ac < bc$ (d) $a = b > 0$
31. For $a, b, c \in R$ if $a > b$ and $c < 0$, then
 (a) $a + c < b + c$ (b) $ac > bc$ (c) $ac < bc$ (d) $a - b < 0$
32. If $a > 0$ and $b < 0$, then
 (a) $ab > 0$ (b) $ab < 0$ (c) $a + b \geq 0$ (d) $a - b < 0$
33. The set $\{1, -1\}$ is closed w. r. t
 (a) Addition (b) Multiplication (c) Subtraction (d) None
34. The set $\{1\}$ has closure property w. r. t
 (a) Addition (b) Subtraction (c) Division (d) None
35. $a(b + c - d) = ab + ac - ad$ is ----- property
 (a) Left distributive (b) Right distributive (c) Associative (d) none
36. If $a < b$ then
 (a) $a < b$ (b) $\frac{1}{a} < \frac{1}{b}$ (c) $\frac{1}{a} > \frac{1}{b}$ (d) $a - b > 0$
37. If $\frac{a}{b} = \frac{ka}{kb}$, $k \neq 0$, this rule is called
 (a) Rules of product of fractions (b) Golden rule of fraction
 (c) Rules of Quotient of fractions (d) principle for equality of fraction
38. If n is an even Integer, then $(i)^n$ is equal to
 (a) i (b) $-i$ (c) ± 1 (d) $\pm i$
39. If n is an odd number then $(i)^n$ is equal to
 (a) i (b) $-i$ (c) ± 1 (d) $\pm i$
40. If n is an integral multiple of 4, then $(i)^n$ is equal to
 (a) 1 (b) -1 (c) ± 1 (d) $\pm i$
41. If $a + ib = c + id$, then it must be true that
 (a) $a = c$ & $b = d$ (b) $a = -c$ & $b = d$
 (c) $a = d$ & $b = c$ (d) $ad = bc$
42. If $a + ib$ is complex number, then its conjugate is
 (a) $a - ib$ (b) $-a - ib$ (c) $\sqrt{a^2 + b^2}$ (d) ab
43. If z is any real number, then its conjugate is
 (a) a real number (b) complex number (c) any Integer (d) zero
44. If k is any real number and $a + ib$ is a complex number, then
 (a) $|k(a + ib)| = ka + ib$ (b) $|k(a + ib)| = ka - ib$
 (c) $|k(a + ib)| = \sqrt{k^2(a^2 + b^2)}$ (d) None of these
45. The additive identity in set of complex numbers is
 (a) (0, 0) (b) (0, 1) (c) (1, 0) (d) (1, 1)

46. The multiplicative Identity of complex numbers is
 (a) (0, 0) (b) (0, 1) (c) (1, 0) (d) (1, 1)
47. The additive Inverse of $(a, -b)$ is
 (a) (a, b) (b) $(a, -b)$ (c) $(-a, -b)$ (d) $(-a, b)$
48. The multiplicative Inverse of $(a, -b)$ is
 (a) $(\frac{a}{a^2+b^2}, \frac{b}{a^2+b^2})$ (b) $(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2})$
 (c) $(\frac{-a}{a^2+b^2}, \frac{b}{a^2+b^2})$ (d) $(\frac{a}{\sqrt{a^2+b^2}}, \frac{-b}{\sqrt{a^2+b^2}})$
49. $(0, 1)$ is equal to
 (a) 1 (b) i (c) $-i$ (d) 0
50. $(0, 1)^2$ is equal to
 (a) 1 (b) -1 (c) i (d) $-i$
51. $(0, 1)^3$ is equal to
 (a) 1 (b) -1 (c) i (d) $-i$
52. $(0, 1)^4$ is equal to
 (a) 1 (b) -1 (c) i (d) $-i$
53. $(-i)^{19}$ is equal to
 (a) i (b) $-i$ (c) 1 (d) -1
54. $(-1)^{\frac{-21}{2}}$ is equal to
 (a) i (b) $-i$ (c) 1 (d) -1
55. $(0, 3)(0, 5)$ is equal to
 (a) 15 (b) -15 (c) $-8i$ (d) $8i$
56. The sum of two conjugate complex numbers is
 (a) a real number (b) an imaginary number
 (c) real or imaginary number (d) not defined
57. The product of two conjugate complex numbers is
 (a) a real number (b) an imaginary number
 (c) May be an Irrational number (d) not defined
58. The multiplicative Inverse of $(-4, 7)$ is
 (a) $(\frac{-4}{65}, \frac{-7}{65})$ (b) $(\frac{4}{65}, \frac{-7}{65})$ (c) $(\frac{-4}{\sqrt{65}}, \frac{-7}{\sqrt{65}})$ (d) $(\frac{4}{\sqrt{65}}, \frac{-7}{\sqrt{65}})$
59. Factors of $3(x^2 + y^2)$ are
 (a) $3(x + y)(x - y)$ (b) $3(x + iy)(x - iy)$
 (c) $\sqrt{3}(x + iy)(x - iy)$ (d) none
60. Real part of $\frac{2+i}{i}$ is equal to
 (a) 1 (b) 2 (c) -1 (d) $\frac{1}{2}$
61. Imaginary part of $(-2 + 3i)^3$ is equal to
 (a) -2 (b) 9 (c) 26 (d) -8

62. If R is the set of real numbers, then product $R \times R$ is called
 (a) Cartesian plane (b) Argand diagram
 (c) Ordered pair (d) real line
63. The geometrical plane on which coordinate system has been specified is called
 (a) Coordinate plane or real plane (b) Argand diagram
 (c) Cartesian plane (d) Real line
64. If a point A of a coordinate plane corresponds to the ordered pair (a, b) , then a and b are called.
 (a) Coordinates of point A (b) Value of point A
 (c) Abscissa of point A (d) ordinates of point A
65. If point A of the coordinate plane corresponds to the ordered pair (a, b) then,
 (a) a is abscissa of point A (b) b is ordinate of point A
 (c) a & b are coordinates of point A (d) all of these
66. The modulus value of a complex number $z = x + iy$ is the distance from
 (a) x -axis (b) y -axis (c) origin (d) (x, y)
67. If $z = x + iy$, then $|z| =$
 (a) $\sqrt{x^2 + y^2}$ (b) $\sqrt{x^2 - y^2}$ (c) $x - iy$ (d) $x^2 + y^2$
68. If $z_1 = 2 + 3i$, $z_2 = 1 - i$ then $|z_1 z_2| =$
 (a) $\sqrt{13}$ (b) $\sqrt{26}$ (c) $\sqrt{15}$ (d) 26
69. The correct statement of De Moivre's Theorem is
 $(\cos \theta + i \sin \theta)^n$ is equal to
 (a) $(\cos \theta + i \sin \theta)^{n+1}$ (b) $(\cos n \theta + i \sin n \theta)$
 (c) $(n \cos \theta + i n \sin \theta)$ (d) $(\cos n \theta - i \sin n \theta)$
70. Polar form of $1 + i\sqrt{3}$ is
 (a) $2(\cos 60^\circ + i \sin 60^\circ)$ (b) $2(\cos 60^\circ - i \sin 60^\circ)$
 (c) $2(\cos 30^\circ + i \sin 30^\circ)$ (d) $\cos 60^\circ + i \sin 60^\circ$
71. Real part of $(x + iy)^n$ is
 (a) $\cos n \theta$ (b) $\sin n \theta$
 (c) $r^n \cos n \theta$ (d) $r^n \sin n \theta$
72. Polar form of $(\sqrt{3} + i)$ is
 (a) $2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$ (b) $2(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})$
 (c) $2(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3})$ (d) $2(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3})$
73. If z is a real number, then
 (a) $\bar{z} = z$ (b) $\bar{z} = |z|$ (c) $z\bar{z} = |z|$ (d) $\bar{z} = -z$
74. If $\bar{z} = 3 - 5i$, then $|z|^2$ is equal to
 (a) 34 (b) $\sqrt{34}$ (c) 16 (d) none

75. If $z = a + ib$, then $(z - \bar{z})^2$ is equal to
 (a) $2a$ (b) $4a$ (c) $-4b^2$ (d) $4a^2$
76. If $Z = a + ib$, then $(Z^2 + \bar{Z}^2)$ is equal to
 (a) $2(a^2 + b^2)$ (b) $2(a^2 - b^2)$ (c) $4iab$ (d) $(a^2 + b^2)^2$
77. If $z = x + iy$, then argument of z is.
 (a) $\theta = \tan^{-1}\left(\frac{x}{y}\right)$ (b) $\theta = \tan^{-1}\left(\frac{y}{x}\right)$
 (c) $\theta = -\tan^{-1}(xy)$ (d) $\theta = -\tan^{-1}\left(\frac{y}{x}\right)$
78. If $|x + 5i| = 3$, then x is equal to
 (a) ± 4 (b) $\pm 4i$ (c) $\pm 22i$ (d) none of these
79. Golden rule of fraction is that for $k \neq 0$ $\frac{a}{b} =$
 (a) $\frac{ka}{kh}$ (b) $\frac{ab}{k}$ (c) $\frac{kb}{ka}$ (d) $\frac{kb}{b}$
80. The set $\{1, -1\}$ possesses closure property w.r.t.
 (a) addition (b) multiplication (c) division (d) subtraction
81. $(-1)^{9/2}$ equals:
 (a) 1 (b) -1 (c) i (d) $-i$
82. The modulus of $\bar{Z}$ is:
 (a) $\sqrt{a^2 - b^2}$ (b) $a^2 + b^2$ (c) $\sqrt{a^2 + b^2}$ (d) $\sqrt{a^2 - (ib)^2}$
83. $0.1428571428571\ldots$ is:
 (a) irrational number (b) rational number
 (c) natural number (d) decimal number
84. i^{13} equals:
 (a) i (b) 1 (c) -1 (d) $-i$

Chapter - 2

Multiple Choice Questions

(Encircle the correct answer choice)

1. If $A \subseteq B$ and $B \subseteq A$ then which is true
 (a) $A = B$ (b) $A \neq B$ (c) $A \cap B = \emptyset$ (d) $A \cup B = \emptyset$
2. If $(1-1)$ correspondence can be established in two sets A and B , then it must be true that
 (a) $A = B$ (b) $A \sim B$ (c) $A \cap B = \emptyset$ (d) $A \cap B \neq \emptyset$
3. The set N of natural numbers and O of odd number are
 (a) $N \sim O$ (b) $N \cap O = \emptyset$ (c) $N \cup O = O$ (d) none of these

4. The set N and Z are
 - (a) Equivalent sets
 - (b) Equal sets
 - (c) Disjoint sets
 - (d) finite sets
5. Which of the following is true
 - (a) $N \subset Z$
 - (b) $Z \subset Q$
 - (c) $Q \subset R$
 - (d) all of these
6. If a set S has m elements, then number of subsets in S are
 - (a) m^2
 - (b) 2^m
 - (c) $2^{m \times m}$
 - (d) m
7. If $A \subseteq B$, then
 - (a) B is super set of A
 - (b) $A \cap B = \phi$
 - (c) $B - A = A - B$
 - (d) $A \cap B \neq \phi$
8. If a set S has no proper subset, and then S will be
 - (a) a singleton set
 - (b) empty set
 - (c) an infinite set
 - (d) not a set
9. If a set S has one proper subset only, then S will be
 - (a) a singleton set
 - (b) empty set
 - (c) an infinite set
 - (d) not a set
10. If a set S has n elements, then number of elements in $P(S) =$
 - (a) n^2
 - (b) 2^n
 - (c) $2^{n \times n}$
 - (d) n
11. The set of all subsets of a set is called
 - (a) Power set
 - (b) Subset
 - (c) Super set
 - (d) Infinite set
12. If $S = \{ \}$, then order of set S is
 - (a) 0
 - (b) 1
 - (c) Infinite set
 - (d) not defined
13. The Power set of an empty set has
 - (a) No elements
 - (b) One element
 - (c) Infinity many elements
 - (d) Two elements
14. If $n(S) = m$, then $n(P(S))$ is equal to
 - (a) m^2
 - (b) 2^m
 - (c) $2^{m \times m}$
 - (d) m
15. The set of all elements under consideration is called
 - (a) Universe of discourse
 - (b) Universe
 - (c) an infinite set
 - (d) Finite set
16. The set of real numbers between 1 and 3 is
 - (a) Infinite set
 - (b) $\{2\}$
 - (c) finite set
 - (d) a group
17. Tabular form of $\{x \mid x \in Q, x = -x\}$ is
 - (a) $\{0\}$
 - (b) $\{ \}$
 - (c) all Rational
 - (d) $\{2\}$
18. Which of the following is true
 - (a) $a \in \{ \{a\} \}$
 - (b) $\phi \in \{a\}$
 - (c) $\phi \subseteq \{ \{a\} \}$
 - (d) $\phi \in \{ \{ \phi \} \}$
19. The set builder form of $A \cup B$ is equal to
 - (a) $\{x \mid x \in A \wedge x \in B\}$
 - (b) $\{x \mid x \in A \vee x \in B\}$
 - (c) $\{x \mid x \in A \wedge x \in B\}$
 - (d) $\{x \mid x \in B \wedge x \in A\}$

20. The set builder form of $A \cap B$ is equal to
 (a) $\{x \mid x \in A \wedge x \in B\}$ (b) $\{x \mid x \in A \vee x \in B\}$
 (c) $\{x \mid x \in A \wedge x \notin B\}$ (d) $\{x \mid x \in B \wedge x \notin A\}$
21. The set builder form of $A - B$ is equal to
 (a) $\{x \mid x \in A \wedge x \in B\}$ (b) $\{x \mid x \in A \vee x \in B\}$
 (c) $\{x \mid x \in A \wedge x \notin B\}$ (d) $\{x \mid x \in B \wedge x \notin A\}$
22. The set builder form of $B - A$ is equal to
 (a) $\{x \mid x \in A \wedge x \in B\}$ (b) $\{x \mid x \in A \vee x \in B\}$
 (c) $\{x \mid x \in A \wedge x \notin B\}$ (d) $\{x \mid x \in B \wedge x \notin A\}$
23. If $A \cap B = \phi$, then A and B are
 (a) Disjoint sets (b) over lapping sets
 (c) Equal sets (d) Equivalent sets
24. If $A \cap B \neq \phi$ then A and B are
 (a) Disjoint sets (b) over lapping sets
 (c) Equal sets (d) Equivalent sets
25. In set builder form A^c is written as
 (a) $\{x \mid x \in U \wedge x \in A\}$ (b) $\{x \mid x \in U \vee x \in A\}$
 (c) $\{x \mid x \in U \wedge x \notin A\}$ (d) $\{x \mid x \in A \wedge x \notin U\}$
26. If a set consists of those elements of A which are not in B, then the set is
 (a) $A \cup B$ (b) $A \cap B$ (c) $A - B$ (d) $B - A$
27. Let A and B are two non empty sets and U be a universal set, then $A - B$
 (a) $A \cap B^c$ (b) $B - A$ (c) $\cup$ (d) ϕ
28. If $A \cap B \neq \phi$ i.e. sets A and B are disjoint, then $n(A \cup B)$ is equal to
 (a) $n(A) + n(B)$ (b) $n(A) \cdot n(B)$
 (c) $n(A) + n(B) - n(A \cap B)$ (d) $n(A \cap B)$
29. If $A \cap B \neq \phi$ i.e. sets A and B are overlapping, then $n(A \cup B)$ is equal to
 (a) $n(A) + n(B)$ (b) $n(A) \cdot n(B)$
 (c) $n(A) + n(B) - n(A \cap B)$ (d) $n(A \cap B)$
30. If $A \subseteq B$, then $n(A \cup B)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A \cap B)$ (d) 0
31. If $B \subseteq A$, then $n(A \cup B)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A \cap B)$ (d) 0
32. If $A \cap B = \phi$, then $n(A \cap B)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A \cap B)$ (d) 0
33. If $A \cap B \neq \phi$ i.e. A and B are overlapping sets, then $n(A \cap B)$
 (a) 0 (b) $n(A) + n(B)$
 (c) $n(A) \cdot n(B)$ (d) cannot be determined
34. If $A \subseteq B$, then $n(A \cap B)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A) + n(B)$ (d) $n(A) \cdot n(B)$

35. If $B \subseteq A$ then, $n(A \cap B)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A) + n(B)$ (d) $n(A) \cdot n(B)$
36. If A and B are Disjoint sets i.e. $A \cap B = \phi$, then $n(A \cup B)$ is equal to
 (a) $n(A)$ (b) $n(B)$
 (c) $n(A) + n(B) - n(A \cup B)$ (d) $n(A) - n(B)$
37. If A and B are disjoint sets i.e. $A \cap B = \phi$, then $n(B - A)$
 (a) $n(A) + n(B)$ (b) $n(A) \cdot n(B)$
 (c) $n(A) + n(B) - n(A \cup B)$ (d) $n(B) - n(A)$
38. If $A \subseteq B$, then $n(A - B)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A \cap B)$ (d) 0
39. If $B \subseteq A$, then $n(B - A)$ is equal to
 (a) $n(A)$ (b) $n(B)$ (c) $n(A \cap B)$ (d) 0
40. If $B \subseteq A$, $A - B \neq \phi$, then $n(A - B)$
 (a) $n(A)$ (b) $n(B)$ (c) $n(A) - n(B)$ (d) 0
41. Which of following is true
 (a) $A \cup \phi = A$ (b) $A \cap \phi = \phi$ (c) $A - \phi = A$ (d) All of these
42. Which of following is true
 (a) $\phi - A = \phi$ (b) $A \cup A = A$
 (c) $A \cap A = A$ (d) $A - A = \phi$ (e) all of these
43. Which of following is true
 (a) $A \cup U = U$ (b) $A - U = \phi$
 (c) $A \cap U = A$ (d) $U - A = A'$ (e) all of these
44. If $A \cup B = A$, then
 (a) $A \subseteq B$ (b) $B \subseteq A$ (c) $A = \phi$ (d) None of these
45. De Morgan's Laws are
 (a) $(A \cup B)' = A' \cap B'$ (b) $(A \cup B)' = A' \cup B'$
 (c) $(A \cap B)' = A' \cup B'$ (d) $(A \cap B)' = (A \cup B)'$
46. De Morgan's Laws are
 (a) $(A \cap B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
 (c) $(A \cap B)^c = A^c \cup B^c$ (d) $(A \cap B)^c = (A \cup B)^c$
47. The way of drawing conclusions from a limited number of observations is called
 (a) An Induction (b) deduction
 (c) proposition (d) postulate
48. The way of drawing conclusions from premises believed to be true is called
 (a) an Induction (b) deduction
 (c) proposition (d) postulate

49. A statement which is accepted to be true without proof and used to find other conclusion is called.
 (a) An Induction (b) deduction
 (c) proposition (d) postulate
50. Logic in which every statement is regarded as true or false is called
 (a) Aristotelian logic (b) Non Aristotelian logic
 (c) Proposition (d) postulate
51. The logic in which there is a scope of more than two possibilities is called.
 (a) Aristotelian logic (b) Non Aristotelian logic
 (c) Proposition (d) postulate
52. A statement which can be decided as true or false is called
 (a) proposition (b) postulate
 (c) compound proposition (d) truth value
53. The symbol which is used to denote negation of a proposition is
 (a) $\sim$ (b) $\rightarrow$ (c) $\wedge$ (d) $\vee$
54. If $p \rightarrow q$ is a conditional, then p is called
 (a) antecedent (b) conclusion (c) consequence (d) conjunction
55. If $p \rightarrow q$ is a Implication, then q is called
 (a) Hypothesis (b) conclusion (c) antecedent (d) converse
56. The symbol which is used to combine propositions is called
 (a) Connective (b) Negation
 (c) operator (d) compound proposition
57. If p and q be two propositions, then $p \wedge q$ is
 (a) Conjunction (b) disjunction
 (c) conditional (d) Bi conditional
58. If p and q be two propositions, then $p \rightarrow q$ is
 (a) Conjunction (b) disjunction
 (c) conditional (d) Bi conditional
59. If p and q be two propositions, then $p \leftrightarrow q$ is
 (a) conjunction (b) disjunction
 (c) conditional (d) Bi conditional
60. A compound proposition which is always true is called
 (a) Tautology (b) contradiction
 (c) Absurdity (d) contingency
61. A compound proposition which is neither always true nor false is called
 (a) Tautology (b) contradiction
 (c) Absurdity (d) contingency
62. A compound proposition which is always wrong is called
 (a) Tautology (b) absurdity
 (c) contingency (d) Equivalence
63. If p be proposition, then $(p \vee \sim p)$ is
 (a) Tautology (b) absurdity (c) contingency (d) Equivalence

64. If P be any proposition then $(p \wedge \sim p)$ is
 (a) Tautology (b) absurdity
 (c) contingency (d) Equivalence
65. If $\sim p \rightarrow q$ is a conditional, then its converse is
 (a) $q \rightarrow \sim p$ (b) $p \rightarrow \sim q$ (c) $\sim q \rightarrow p$ (d) $\sim q \rightarrow \sim p$
66. If $\sim p \rightarrow q$ is a conditional then its inverse is
 (a) $q \rightarrow \sim p$ (b) $p \rightarrow \sim q$ (c) $\sim q \rightarrow p$ (d) $\sim q \rightarrow \sim p$
67. If $\sim p \rightarrow q$ is a conditional then its contra positive is
 (a) $q \rightarrow \sim p$ (b) $p \rightarrow \sim q$ (c) $\sim q \rightarrow p$ (d) $\sim q \rightarrow \sim p$
68. If p is a proposition $4 < 5$, q is a proposition $2 + 5 = 8$, then truth value of $p \wedge q$ is
 (a) T (b) F (c) Neither T nor F (d) Either T or F
69. If p is a proposition $4 < 5$, q is a proposition $2 + 5 = 8$, then truth value of $p \vee q$ is
 (a) T (b) F (c) Nether T for F (d) Either T or F
70. If p is a proposition $4 < 5$, q is a proposition $2 + 5 > 8$, then truth value of $p \rightarrow q$ is
 (a) T (b) F (c) Nether T for F (d) Either T or F
71. If p is a proposition $4 < 5$, q is a proposition $2 + 5 \neq 8$, then truth value of $p \leftrightarrow q$ is
 (a) T (b) F (c) Nether T for F (d) Either T or F
72. For the propositions p and q , $(p \wedge q) \rightarrow p$ is
 (a) Tautology (b) Absurdity
 (c) contingency (d) none of these
73. For the propositions p and q , $p \rightarrow (p \vee q)$ is
 (a) Tautology (b) Absurdity
 (c) contingency (d) none of these
74. The words or symbols which convey the idea of quantity or number is called
 (a) Quantifier (b) Negation
 (c) conditional (d) Truth table
75. The symbol which is used to convey the idea of all objects under consideration is called
 (a) Universal quantifier (b) Existential Quantifier
 (c) Universal set (d) Non of these
76. The logical form of $(A \cap B)' = A' \cup B'$ is
 (a) $\sim (p \wedge q) = \sim p \wedge \sim q$ (b) $\sim (p \wedge q) = \sim p \vee \sim q$
 (c) $\sim (p \vee q) = \sim p \vee \sim q$ (d) $\sim (p \vee q) = \sim p \wedge \sim q$
77. The logical form of $(A \cup B)^c = A^c \cap B^c$ is
 (a) $\sim (p \wedge q) = \sim p \wedge \sim q$ (b) $\sim (p \wedge q) = \sim p \vee \sim q$
 (c) $\sim (p \vee q) = \sim p \vee \sim q$ (d) $\sim (p \vee q) = \sim p \wedge \sim q$

78. If p and q are two propositions, then truth set of $p \vee q$ is
(a) $P \cap Q$ (b) $P \cup Q$ (c) $P - Q$ (d) $Q - P$
79. If p and q are two propositions then truth set of $p \wedge q$ is
(a) $P \cap Q$ (b) $P \cup Q$ (c) $P - Q$ (d) $Q - P$
80. If p and q be two propositions, then truth set of $p \rightarrow q$ is
(a) $P' \cup Q$ (b) $P' \cap Q$ (c) $P = Q$ (d) $P \cap Q'$
81. Truth set of $p \leftrightarrow q$ is
(a) $P' \cap Q'$ (b) $P' \cup Q'$ (c) $P = Q$ (d) $P \cup Q$
82. If p is a proposition, then truth set of $\sim p$ is
(a) P' (b) $\cup$ (c) ϕ (d) None
83. Truth set of a tautology is
(a) Universal set (b) ϕ (c) True (d) False
84. Truth set of a contradiction is
(a) Universal set (b) ϕ (c) True (d) False
85. Logical form of $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ is
(a) $p \vee (q \wedge r) = (p \vee q) \wedge r$ (b) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
(c) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ (d) $p \wedge (q \vee r) = (p \wedge q) \vee r$
86. If set A has 2 elements and B has 4 elements, then number of elements in $A \times B$ is
(a) 6 (b) 8 (c) 16 (d) None of These
87. Every subset of Cartesian product $A \times B$ is called
(a) Relation (b) Function (c) Domain (d) Range
88. The empty set $\{ \}$ being the subset of $A \times B$ is
(a) Binary relation. (b) Function
(c) Ordered pair (d) None of these
89. If $f: A \rightarrow B$ be a function, then it is an into function if
(a) $\text{Range} = B$ (b) $\text{Range} \subset B$
(c) Range is not repeated (d) $\text{Domain} \neq A$
90. A function $f: A \rightarrow B$ is called an on to if
(a) $\text{Domain} \subset A$ (b) $\text{Range} \subset B$
(c) $\text{Range} = B$ (d) $\text{Domain} \sim \text{Range}$
91. A function $f: A \rightarrow B$ is (1 - 1) if
(a) $\text{Domain} \subset A$ (b) $\text{Range} \subset B$
(c) $\text{Domain} = \text{Range}$ (d) Range is not repeated
92. A function $f: A \rightarrow B$ is (1 - 1) and onto if
(a) $\text{Domain} = A$ (b) $\text{Range} \subseteq B$
(c) $\text{Domain} = \text{Range}$ (d) $\text{Range} = B$ and Range is not repeated
93. A (1 - 1) function is also called Function
(a) Injective (b) Surjective (c) Bijective (d) Inverse
94. An onto function is also called Function
(a) Injective (b) Surjective (c) Bijective (d) Inverse

95. A (1 - 1) and on to function is also called Function
(a) Injective (b) Surjective (c) Bijective (d) Inverse
96. Inverse of a function Exists only if it is
(a) Injective (b) Bijective (c) Surjective (d) all of these
97. The function $f = \{(x, y) \mid y = mx + c\}$, m & c are real number is
(a) Linear (b) Quadratic (c) A circle (d) A point
98. The function $f = \{(x, y) \mid y = ax^2 + bx + c, a \neq 0\}$ is
(a) Linear (b) Quadratic (c) A circle (d) A point
99. Inverse of line is
(a) a line (b) a parabola (c) a point (d) not defined
100. If $y = \sqrt{x}$, $x \geq 0$ is a function, then its inverse is
(a) a line (b) a parabola (c) a point (d) not a function
101. The function $f = \{(x, y) \mid y = x\}$ is
(a) Identity function (b) Null function
(c) not a function (d) similar function
102. If a set A has 2 elements and B has 3 elements, then different relations in $A \times B$ are
(a) 5 (b) 6 (c) 8 (d) 64
103. If a set A has 2 elements and B has 3 elements, then different function in $A \times B$ are
(a) 6 (b) 8 (c) 9 (d) not defined
104. If a set A has m elements and B has n elements, than relations in $A \times B$
(a) $m \times n$ (b) $2^{m \times n}$ (c) $m + n$ (d) $(m \times n)^2$
105. If a set S has n elements, then different relations is A
(a) $2n$ (b) 2^{2n} (c) n^2 (d) 2^n
106. The Inverse function of $\{(x, y) \mid y = mx + c\}$ is
(a) $\{(y, x) \mid x = my + c\}$ (b) $\{(x, y) \mid x = my + c\}$
(c) $\{(y, x) \mid y = mx\}$ (d) not a function
107. An operation which is performed on a single number is called
(a) Unary operation (b) Binary operation
(c) Relation (d) function
108. Squaring a number is
(a) unary operation (b) Binary operation
(c) relation (d) function
109. Which of the following is not a binary operation
(a) + (b) $\div$ (c) $\sqrt{\quad}$ (d) -
110. For a non empty set G, a function from $G \times G \rightarrow G$ is called
(a) Binary operation (b) Unary operation
(c) Groupied (d) Binary relation

111. Any subset of $G \times G$ is called
 (a) Binary operation (b) relation
 (c) function (d) Cartesian product
112. The set $\{1, -1, i, -i\}$ is closed w. r. t
 (a) + (b) - (c) $\times$ (d) *
113. The set of odd number is not closed w. r. t
 (a) + (b) $\times$ (c) - (d) + & -
114. Let S be a not empty set and * is binary operation in it. If closure property holds in S, then S is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
115. If N is set of natural number, then $(N, +)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
116. If W is the set of whole numbers, than $(W, +)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
117. If N is set of natural number, then $(N, \times)$ or $(N, \cdot)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
118. For a non empty sets S, $(P(S), \cap)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
119. For a non empty sets S, $(P(S), \cup)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
120. If Z is set of Integers, than $(Z, \cdot)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
121. If R is the set of real numbers, then $(R, +)$ is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
122. If Q is the set of rational numbers, than $(Q, \cdot)$
 (a) Groupied (b) Semi group (c) Monoid (d) Group
123. If S is non empty set. Then identity element in $P(S)$, w. r. t $\cap$
 (a) $\{\}$ (b) S (c) $\{\phi\}$ (d) does not exist
124. If S is non empty set. Than identity element in $P(S)$, w. r. t $\cup$
 (a) $\{\}$ (b) S (c) $\{\phi\}$ (d) does not exist
125. The set of non -zero real numbers w. r. t multiplication is
 (a) Groupied (b) Semi group (c) Monoid (d) Group
126. Identity element in $(C, +)$ is
 (a) (0, 0) (b) (0, 1) (c) (1, 0) (d) (1, 1)
127. Identity element in $(C, \cdot)$ is
 (a) (0, 0) (b) (0, 1) (c) (1, 0) (d) (1, 1)
128. The set of first elements of ordered pairs in a relation is called its:
 (a) domain (b) range (c) co-domain (d) relation
129. If A and B are disjoint sets then :
 (a) $A \cup B = \phi$ (b) $A \cap B = \phi$ (c) $A \subset B$ (d) $A - B = \phi$
130. If $S = \{1, 2, 3, 4, 5, 6\}$ then $n(S)$ equals:
 (a) 2^6 (b) 6 (c) $6!$ (d) - 6

131. If $A = \phi$, then $P(A)$ is:
 (a) Empty set (b) $\{0\}$ (c) $\{\phi\}$ (d) none of these
132. The graph of linear function is:
 (a) circle (b) straight line (c) parabola (d) triangle
133. A system of linear equations involves at least... equation(s):
 (a) 1 (b) 2 (c) 3 (d) 4
134. If $A \subseteq B$, then $A \cap B$ is equal to:
 (a) ϕ (b) A (c) B (d) $-A$
135. If $A = \{1, 2, 3\}$, $B = \{3, 4\}$, then $A - B$ is:
 (a) $\{4\}$ (b) $\{1, 2\}$ (c) $\{1, 4\}$ (d) $\{3\}$
136. The number of elements in a set B is 4, the number of elements in $P(B)$
 (a) 16 (b) 12 (c) 8 (d) 4
137. The number of all subsets of a set having three elements is:
 (a) 4 (b) 6 (c) 8 (d) 10
138. Set of all possible sub sets of a set S is called:
 (a) equivalent set (b) empty set (c) power set (d) sub set
139. Set of integers is a group w.r.t:
 (a) addition (b) multiplication (c) subtraction (d) division
140. f is function from A to B . Domain of f is equal to:
 (a) any subset of A (b) $A \times B$ (c) A (d) B
141. Every function is a:
 (a) relation (b) inverse function
 (c) one to one (d) none of these
142. Inverse of any element of a group is:
 (a) not unique (b) unique
 (c) has many inverses (d) none of these

Chapter - 3

Multiple Choice Questions

(Encircle the correct answer choice)

- A rectangular array of numbers enclosed by a pair of brackets is called a
 (a) matrix (b) Row (c) column (d) determinant
- The horizontal lines of numbers in a matrix are called
 (a) Rows (b) column (c) column matrix (d) Row matrix
- The vertical lines of numbers in a matrix are called
 (a) Rows (b) columns (c) column matrix (d) Row matrix
- If a matrix A has m rows and n column, then order of A is
 (a) $m \times n$ (b) $n \times m$ (c) $m + n$ (d) m^n
- The element a_{ij} of any matrix A is present in
 (a) i^{th} row and j^{th} column (b) i^{th} column and j^{th} row
 (c) $(i + j)^{\text{th}}$ row and column (d) $(i - j)^{\text{th}}$ row and column

6. Any matrix A is called real if all a_{ij} are
(a) real numbers (b) Imaginary numbers (c) 0 (d) 1
7. If any matrix A has only one row, then it is called
(a) row matrix (b) column matrix
(c) Square matrix (d) Rectangular matrix
8. If any matrix A has only one column, then it is called
(a) row matrix (b) column matrix
(c) Square matrix (d) Rectangular matrix
9. If a matrix A has same numbers of rows and column, then A is called
(a) row matrix (b) column matrix
(c) Square matrix (d) Rectangular matrix
10. If any matrix A has different numbers of rows and column, then A is
(a) row matrix (b) column matrix
(c) Square matrix (d) Rectangular matrix
11. Any matrix of order $m \times 1$ is called
(a) row matrix (b) column matrix
(c) Square matrix (d) Rectangular matrix
12. Any matrix of order $1 \times n$ is called
(a) row matrix (b) column matrix
(c) Square matrix (d) Rectangular matrix
13. For the square matrix $A = [a_{ij}]_{n \times n}$, the elements
 $a_{11}, a_{22}, a_{33}, \dots, a_{nn}$ are
(a) principal diagonal or leading diagonal (b) Secondary diagonal
(c) central row (d) central column
14. For the matrix $A = [a_{ij}]_{n \times n}$, the elements
 $a_{1n}, a_{2n-1}, a_{3n-2}, a_{4n-3}, \dots, a_{n1}$ form
(a) Main diagonal (b) Leading diagonal
(c) principal diagonal (d) Secondary diagonal
15. For the square matrix $A = [a_{ij}]$. If all $a_{ij} = 0, i \neq j$ and at least one
 $a_{ij} \neq 0, i = j$, then A is called
(a) Diagonal matrix (b) Scalar matrix
(c) Identity matrix (d) Null matrix
16. For the square matrix $A = [a_{ij}]$. If all $a_{ij} = 0, i \neq j$ and all $a_{ij} = k$
(non zero) for $i = j$, then A is called
(a) Diagonal matrix (b) Scalar matrix
(c) Identity matrix (d) Null matrix
17. If all off diagonal elements are zeros and at least one of the
leading diagonal is non zero, then matrix is called
(a) Diagonal matrix (b) Scalar matrix
(c) Identity matrix (d) Null matrix
18. The matrix $[7]$ is
(a) square matrix (b) Row matrix
(c) column matrix (d) all of these

19. If A is a matrix of order $m \times n$, then the matrix of order $n \times m$ is called
 (a) Transpose of A (b) Inverse of A
 (c) Main diagonal of A (d) Echelon form of A
20. Two matrix A and B are said to be conformable for addition if
 (a) number of columns in A = number of rows in B
 (b) number of rows in B = number of columns in A
 (c) rows of A = columns of B (d) order of A = order of B
21. If $[a_{ij}] = A$, and $[b_{ij}] = B$, then $A = B$ if and only if
 (a) order of A = order of B (b) $a_{ij} = b_{ij}$, $i = j$ only
 (c) $a_{ij} = b_{ij}$, ($i \neq j$ only) (d) $a_{ij} = b_{ij}$ for all i & j
22. For any two matrices A and B , $(A + B)^t$ is equal to
 (a) $A^t + B^t$ (b) $(A+B)$ (c) $A^t B^t$ (d) $B^t A^t$
23. $(AB)^t$ is equal to
 (a) $B^t A^t$ (b) $A^t B^t$ (c) $A B$ (d) $(B A)^t$
24. $(kAB)^t =$
 (a) $k A^t B^t$ (b) $k B^t A^t$ (c) $k(BA)^t$ (d) $k^t (AB)$
25. Let A be any matrix and n is an Integer, then $A + A + A + \dots +$ to n terms
 (a) A^n (b) $n A$ (c) A^{n-1} (d) $(n+1) A$
26. Two matrix A and B are conformable for multiplication AB if
 (a) number of columns in A = number of rows in B
 (b) number of rows in B = number of columns in A
 (c) number of rows in A = number of rows in B
 (d) number of columns in A = number of columns in B
27. If A is a matrix of order $m \times n$ and B of order $n \times q$, then order of AB is
 (a) $m \times q$ (b) $n \times n$ (c) $m \times m$ (d) $q \times m$
28. If A is of order 2×3 and B of order 4×2 , then order of AB
 (a) 2×2 (b) 3×4 (c) 4×3 (d) Non
29. If A is of order 2×3 and B of order 4×2 , then order of BA
 (a) 2×2 (b) 3×4 (c) 4×3 (d) Non
30. If $AB = BA$, then which is true
 (a) A and B are multiplicative inverse of each other
 (b) One of A or B is null matrix
 (c) One of A or B is identity matrix (d) all of these
31. For any square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $|A|$ is equal to
 (a) $ab - cd$ (b) $ad - bc$ (c) $ac - bd$ (d) $bc - ad$
32. If $A = [-7]$, then $|A|$ is equal to
 (a) 7 (b) -7 (c) 0 (d) Not possible

33. If A is any square matrix of order 3, then $|kA|$ is equal to
 (a) $|k||A|$ (b) $|k||A|$ (c) $k^2|A|$ (d) $k^3|A|$
34. If A is any square matrix and $AB = BA = I$, then B is called
 (a) Additive Inverse of A (b) Multiplicative Inverse of A
 (c) Transpose of A (d) determinant of A
35. If $A + B = B + A = O$, then B is called
 (a) Additive Inverse of A (b) Multiplicative Inverse of A
 (c) Transpose of A (d) determinant of A
36. If adjoint of $A = \begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}$, Then matrix $A =$
 (a) $\begin{bmatrix} -1 & -2 \\ 4 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}$ (c) $\begin{bmatrix} -4 & 3 \\ -2 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & 2 \\ -3 & -1 \end{bmatrix}$
37. If A is a non-singular matrix, then A^{-1}
 (a) $\frac{1}{|A|} \text{Adj. } A$ (b) $\frac{-1}{|A|} \text{adj. } A$ (c) $\frac{|A|}{\text{adj. } A}$ (d) $\frac{1}{|A| \text{adj. } A}$
38. If $AX = B$, then X is equal to
 (a) $A^{-1}B$ (b) $\frac{B}{A}$ (c) BA^{-1} (d) all of these
39. Inverse of a matrix exist if it is
 (a) Singular (b) Non-singular
 (c) Null matrix (d) Rectangular matrix
40. Which of the property does not hold in matrix multiplication
 (a) Associative (b) Commutative (c) Closure (d) None
41. Let $A = [a_{ij}]$ be a square matrix and M_{ij} is the determinant obtained by deleting i^{th} row and j^{th} column of A . Then Minor of a_{ij} is equal to
 (a) M_{ij} (b) $(-1)^{i+j} M_{ij}$ (c) $(-1)^{i-j} M_{ij}$ (d) $(-1)^{i+j} a_{ij}$
42. Let $A = [a_{ij}]$ be a square matrix and M_{ij} is the determinant obtained by deleting i^{th} row and j^{th} column of A . Then cofactor of a_{ij} is equal to
 (a) M_{ij} (b) $(-1)^{i+j} M_{ij}$ (c) $(-1)^{i-j} M_{ij}$ (d) $(-1)^{i+j} a_{ij}$
43. For any square matrix A , It is always true that.
 (A) $A = A^t$ (b) $-A = \bar{A}$ (c) $|A| = |A^t|$ (d) $A^{-1} = \frac{1}{A}$
44. For any triangular matrix A , $|A|$ is equal to
 (a) Product of leading diagonal elements
 (b) Sum of leading diagonal elements
 (c) Product of secondary diagonal elements
 (d) Product of both diagonal elements

45. If all entries of a square matrix of order 3 is multiplied by k , then value of $|kA|$ is equal to
 (a) $|k| |A|$ (b) $k |A|$ (c) $|A|$ (d) $k^3 |A|$
46. For any non singular matrix A , It is true that
 (a) $A^{-1} = A$ (b) $|A| = A$ (c) $(A^{-1})^t = (A^t)^{-1}$ (d) Non
47. For any non singular matrix A , It is true that
 (a) $(A^{-1})^{-1} = A$ (b) $(A^t)^t = A$ (c) $\overline{\overline{A}} = A$ (d) all of these
48. For any non-singular Matrices A and B it is true that
 (a) $(AB)^{-1} = B^{-1} A^{-1}$ (b) $(AB)^t = B^t A^t$
 (c) $AB = BA$ (d) all of these
49. A square matrix $A = [a_{ij}]$ for which all $a_{ij} = 0, i > j$, then A is called
 (a) Upper Triangular (b) lower Triangular
 (c) Symmetric (d) Hermitian
50. A square matrix $A = [a_{ij}]$ for which all $a_{ij} = 0, i < j$, then A is called
 (a) Upper Triangular (b) lower Triangular
 (c) Symmetric (d) Hermitian
51. A triangular matrix is always a
 (a) Diagonal matrix (b) Scalar matrix
 (c) Square matrix (d) all of these
52. Any square matrix A is called a singular is
 (a) $|A| = 0$ (b) $|A| \neq 0$ (c) $A^t = A$ (d) $AA^{-1} = I$
53. A non empty set F is called field if
 (a) F is a an abelian group under '+'
 (b) $F - \{0\}$ is an abelian group under ' $\cdot$ '
 (c) Right distributive property holds (d) all of these
54. Which of the following sets is a field
 (a) R (b) Q (c) C (d) all of these
55. Which of the following sets is not a field
 (a) R (b) Q (c) C (d) Z
56. The system of linear Equations involving the same variables are equivalent if they have
 (a) Number of equations = number of variables (b) same solutions
 (c) different solutions (d) infinity many solutions
57. A square matrix A is symmetric if
 (a) $A^t = A$ (b) $A^t = -A$ (c) $(\overline{A})^t = A$ (d) $(\overline{A})^t = -A$
58. A square matrix A is skew symmetric if
 (a) $A^t = A$ (b) $A^t = -A$ (c) $(\overline{A})^t = A$ (d) $(\overline{A})^t = -A$
59. A square matrix A is Hermitian if
 (a) $A^t = A$ (b) $A^t = -A$ (c) $(\overline{A})^t = A$ (d) $(\overline{A})^t = -A$
60. A square matrix A is skew Hermitian if
 (a) $A^t = A$ (b) $A^t = -A$ (c) $(\overline{A})^t = A$ (d) $(\overline{A})^t = -A$

61. The main diagonal elements of a skew symmetric matrix must be
(a) 1 (b) 0
(c) any non zero number (d) any complex number
62. The main diagonal elements of a skew Hermitian matrix must be
(a) 1 (b) 0
(c) any non zero number (d) any complex number
63. In echelon form of a matrix, the first non zero entry is called
(a) leading entry (b) first entry
(c) Preceding entry (d) Diagonal entry
64. The additive inverse of a matrix exist only if it is
(a) singular (b) non singular
(c) null matrix (d) any matrix of order $m \times n$
65. The multiplicative inverse of a matrix exist only if it is
(a) singular (b) non singular
(c) null matrix (d) any matrix of order $m \times n$
66. If $\begin{vmatrix} a & b \\ 0 & 7 \end{vmatrix} = \begin{vmatrix} 2 & 3 \\ 1 & -9 \end{vmatrix}$ then
(a) $a = -3$ (b) $a = b$ (c) $a = \frac{1}{3}$ (d) $a = \frac{-1}{3}$
67. The number of non zero rows in echelon form of a matrix is called
(a) order of a matrix (b) Rank of a matrix
(c) leading (d) leading row
68. If A is any square matrix then $A + A^t$ is a
(a) Symmetric (b) skew symmetric
(c) Hermitian (d) skew hermitian
69. If A is any square matrix then $A - A^t$ is a
(a) Symmetric (b) skew symmetric
(c) Hermitian (d) skew hermitian
70. If A is any square matrix then $A + (\bar{A})^t$ is a
(a) Symmetric (b) skew symmetric
(c) Hermitian (d) skew hermitian
71. If A is any square matrix then $A - (\bar{A})^t$ is a
(a) Symmetric (b) skew symmetric
(c) Hermitian (d) skew hermitian
72. If A is symmetric (skew symmetric), than A^2 must be
(a) singular (b) non- singular
(c) Symmetric (d) Anti symmetric
73. In a homogeneous system of linear equations, The solution (0,0,0) is
(a) Trivial solution (b) non trivial solution
(c) exact solution (d) Non
74. If $AX = O$, then $X =$
(a) I (b) O (c) A^{-1} (d) not possible

75. If a system of linear equations have no solution at all, then it is called a/an
 (a) Consistent system (b) Inconsistent system
 (c) Trivial system (d) Non Trivial system
76. The value of λ for which the system $x + 2y = 4$; $2x + \lambda y = -3$ does not possess the unique solution
 (a) 4 (b) -4 (c) ± 4 (d) any real number
77. If the system $x + 2y = 0$; $2x + \lambda y = 0$ has non trivial solution, then λ is
 (a) 4 (b) -4 (c) ± 4 (d) any real number
78. If $\begin{bmatrix} 2x+3 & 1 \\ -3 & 4 \end{bmatrix} = \begin{bmatrix} -1+x & 1 \\ -3 & 4 \end{bmatrix}$, then $x =$
 (a) 3 (b) -3 (c) 4 (d) -4
79. The cofactor A_{22} of $\begin{bmatrix} 1 & 2 & 4 \\ -1 & 2 & 5 \\ 0 & 1 & -1 \end{bmatrix}$ is
 (a) 0 (b) -1 (c) 1 (d) 2
80. If $A = [a_{ij}]_{3 \times 3}$, then $I_3 A$ is equal to:
 (a) A (b) A^{-1} (c) Not possible (d) $-A$
81. If all the entries of a row of a square matrix A are zero, then $|A|$ equals:
 (a) 1 (b) -1 (c) 0 (d) $-|A|$
82. If $\begin{vmatrix} x & 4 \\ 5 & 10 \end{vmatrix} = 0 \Rightarrow x$ equals:
 (a) 2 (b) 4 (c) 6 (d) 8
83. The inverse of unit matrix is:
 (a) unit (b) singular (c) skew symmetric (d) rectangular
84. Transpose of a row matrix is:
 (a) diagonal matrix (b) zero matrix
 (c) column matrix (d) scalar matrix

Chapter - 4

Multiple Choice Questions

(Encircle the correct answer choice)

1. The equation $ax^2 + bx + c = 0$ will be quadratic if
 (a) $a = 0, b \neq 0$ (b) $a \neq 0$
 (c) $a = b = 0$ (d) $b = \text{any real number}$
2. solution set of the equation $x^2 - 4x + 4 = 0$ is
 (a) $\{2, -2\}$ (b) $\{2\}$ (c) $\{-2\}$ (d) $\{4, -4\}$

3. The quadratic formula for solving the equation $ax^2 + bx + c = 0$ is ($a \neq 0$)
- (a) $x = -b \pm \frac{\sqrt{b^2 - 4ac}}{2a}$ (b) $x = -b \pm \sqrt{\frac{b^2 - 4ac}{2a}}$
 (c) $x = -b \pm \frac{\sqrt{b^2 + 4ac}}{2a}$ (d) $x = b \pm \sqrt{\frac{b^2 + 4ac}{2a}}$
4. To convert $ax^{2n} + bx^n + c = 0$ ($a \neq 0$) into quadratic form, the correct substitution
- (a) $y = x^n$ (b) $y = x^{2n}$ (c) $y = x^{-n}$ (d) $y = 1/x$
5. The equation in which variable quantity occurs in exponent is called
- (a) Exponential function (b) Exponential equation
 (c) Reciprocal equation (d) quadratic equation
6. To convert $4^{1-x} + 4^{1+x} = 10$ into quadratic, the substitution is
- (a) $y = 4^{1-x}$ (b) $y = 4^{1+x}$ (c) $y = 4^x$ (d) $y = 4^{-x}$
7. The equation which remains unchanged if x is replaced by $\frac{1}{x}$, then it is called
- (a) Exponential function (b) Exponential equation
 (c) Reciprocal equation (d) quadratic equation
8. The equations involving radical expressions of the variable are called
- (a) reciprocal equations (b) radical equations
 (c) Quadratic equations (d) exponential equations
9. The roots which satisfy radical free equation but not radical equation are called
- (a) Extraneous roots (b) radical roots
 (c) original roots (d) exact roots
10. The cube roots of unity are
- (a) $1, \frac{-1+i\sqrt{3}}{2}, \frac{1+i\sqrt{3}}{2}$ (b) $1, \frac{-1-i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
 (c) $1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$ (d) $1, \frac{-1-i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
11. The cube roots of -1 are
- (a) $-1, \frac{-1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$ (b) $-1, \frac{-1-i\sqrt{3}}{2}, \frac{1+i\sqrt{3}}{2}$
 (c) $-1, \frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$ (d) $-1, \frac{-1-i\sqrt{3}}{2}, \frac{-1+i\sqrt{3}}{2}$
12. Sum of all cube roots of 64, is
- (a) 0 (b) 1 (c) 64 (d) -64
13. Product of all cube roots of -1 is
- (a) 0 (b) 1 (c) 1 (d) Non

14. $16\omega^4 + 16\omega^8 =$
 (a) 0 (b) -16 (c) 16 (d) -1
15. $(-1+\sqrt{-3})^5 + (-1-\sqrt{-3})^5$ is equal to
 (a) 0 (b) 32 (c) -32 (d) -1
16. The sum of all four forth roots of unity is
 (a) unity (b) 0 (c) -1 (d) Non
17. The product of all four forth roots of unity is
 (a) unity (b) 0 (c) -1 (d) Non
18. The sum of all four forth roots are 16 is
 (a) 16 (b) -16 (c) 0 (d) 1
19. The Product of all four forth roots of 81 is
 (a) -81 (b) 81 (c) 0 (d) 1
20. The complex cube roots of unity are each other
 (a) Additive inverse of (b) Equal to each other
 (c) Conjugate of each other (d) Non of these
21. The complex cube roots of unity are each other
 (a) Multiplicative inverse of each other (b) Reciprocal of each other
 (c) Square of each other (d) all of these
22. The complex forth roots of unity are ... each other
 (a) Additive inverse (b) equal to (c) square of (d) Non
23. If sum of all cube roots unity is equal to $x^2 + 1$, than x is equal to
 (a) -1 (b) 0 (c) $\pm i$ (d) 1
24. If product of all cube roots of unity is equal to $p^2 + 1$, then p is
 (a) -1 (b) 0 (c) $\pm i$ (d) 1
25. The complex forth roots of unity are each other
 (a) Multiplicative Inverse (b) complex conjugate
 (c) Additive inverse (d) all of these
26. The expression $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, $a_n \neq 0$ is a polynomial of degree n , if n is any
 (a) Integer (b) non-negative integer
 (c) Positive Integer (d) Real number
27. The expression $x^2 + \frac{1}{x} - 3$ is
 (a) polynomial of degree 2 (b) polynomial of degree 3
 (c) polynomial of degree 1 (d) not a polynomial
28. If $f(x)$ is divided by $x-a$, then Divided = (Divisor) (....) + Remainder
 (a) Divisor (b) Dividend (c) Quotient (d) $f(a)$
29. If $f(x)$ is divided by $x-a$, then by remainder theorem, Remainder is
 (a) $f(a)$ (b) $f(-a)$ (c) $f(a) + R$ (d) $x-a = R$
30. The polynomial $(x-a)$ is a factor of $f(x)$ if and only if
 (a) $f(a) = 0$ (b) $f(a) = R$ (c) quotient = R (d) $x = -a$

31. $x - 2$ is a factor of $x^2 - kx + 4$, if k is
 (a) 2 (b) 4 (c) $k = 8$ (d) -4
32. If $x = -2$ is a root of $kx^2 - 13x^2 + 36 = 0$, then $k =$
 (a) 2 (b) -2 (c) 1 (d) -1
33. $x + a$ is a factor of $x^n + a^n$ when n is
 (a) any integer (b) any positive integer
 (c) any odd integer (d) any real number
34. $x - a$ is a factor of $x^n - a^n$, if n is
 (a) any integer (b) any positive integer
 (c) any odd integer (d) any real number
35. Sum of roots of $ax^2 - bx - c = 0$ is ($a \neq 0$)
 (a) $\frac{b}{a}$ (b) $-\frac{b}{a}$ (c) $\frac{c}{a}$ (d) $-\frac{c}{a}$
36. Product of $ax^2 - bx - c = 0$ is ($a \neq 0$)
 (a) $\frac{b}{a}$ (b) $-\frac{b}{a}$ (c) $\frac{c}{a}$ (d) $-\frac{c}{a}$
37. sum of roots of any quadratic quadratic is
 (a) $\frac{\text{coefficient of } x^2}{\text{coefficient of } x}$ (b) $\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$
 (c) $-\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$ (d) $\frac{\text{constant term}}{\text{coefficient of } x^2}$
38. Product of roots of any quadratic quadratic is
 (a) $\frac{\text{coefficient of } x^2}{\text{coefficient of } x}$ (b) $\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$
 (c) $-\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$ (d) $\frac{\text{constant term}}{\text{coefficient of } x^2}$
39. If sum of roots of $7x^2 + px - q = 0$ is 7, then $p =$
 (a) 7 (b) 49 (c) -49 (d) q
40. If product of roots of $7x^2 - px + q = 0$ is 1, then $q =$
 (a) 7 (b) -7 (c) P (d) 49
41. If 2 and -5 are roots of a quadratic equation, then equation is
 (a) $x^2 - 3x - 10 = 0$ (b) $x^2 - 3x + 10 = 0$
 (c) $x^2 + 3x - 10 = 0$ (d) $x^2 + 3x + 10 = 0$
42. If S and P are sum and product of roots of a quadratic equation, then equation is
 (a) $x^2 - Sx + P = 0$ (b) $x^2 + Sx + P = 0$
 (c) $x^2 + Sx - P = 0$ (d) $x^2 - Sx - P = 0$
43. If α and β the roots of $3x^2 - 2x + 4 = 0$, then value of $\alpha + \beta$
 (a) $\frac{2}{3}$ (b) $-\frac{2}{3}$ (c) $\frac{4}{3}$ (d) $-\frac{4}{3}$

44. If p and q are the roots of $8x^2 - 3x - 16 = 0$ then pq is equal to
(a) 2 (b) -2 (c) $p + q$ (d) None
45. If $ax^2 + bx + c = 0$, then discriminant is
(a) $\sqrt{b^2 - 4ac}$ (b) $\sqrt{b^2 + 4ac}$ (c) $b^2 - 4ac$ (d) $b^2 + 4ac$
46. If roots of $ax^2 + bx + c = 0$, ($a \neq 0$) are real, then
(a) $b^2 - 4ac \geq 0$ (b) $b^2 - 4ac < 0$
(c) $b^2 - 4ac \neq 0$ (d) $b^2 - 4ac \leq 0$
47. The roots of $ax^2 + bx + c = 0$ are imaginary, if
(a) $b^2 - 4ac > 0$ (b) $b^2 - 4ac < 0$
(c) $b^2 - 4ac = 0$ (d) $b^2 - 4ac \neq 0$
48. The roots of $ax^2 + bx + c = 0$ are equal, if
(a) $b^2 - 4ac > 0$ (b) $b^2 - 4ac < 0$
(c) $b^2 - 4ac = 0$ (d) $b^2 + 4ac = 0$
49. If discriminant is positive and perfect square, then roots are
(a) Real & distinct (b) Imaginary & distinct
(c) Rational & distinct (d) irrational and distinct
50. If discriminant is positive and not perfect square, then roots are
(a) Real & distinct (b) Imaginary & distinct
(c) Rational & distinct (d) irrational and distinct
51. If discriminant is negative, then roots are
(a) Real & distinct (b) Imaginary & distinct
(c) Rational & distinct (d) irrational and distinct
52. If discriminant is zero, then roots are
(a) Real & distinct (b) Real & equal
(c) Rational & unequal (d) None of these
53. The roots of $2x^2 - bx + 8 = 0$ are imaginary, if
(a) $b^2 < 64$ (b) $b^2 > 64$ (c) $b^2 = 64$ (d) $b = \pm 8$
54. The equation of the form $ax^2 + bx + c = 0$ where $a, b, c \in R$, $a \neq 0$, is called
(a) Reciprocal equation (b) Quadratic equation
(c) Exponential equation (d) polynomial expression
55. Quadratic equation is also called
(a) 2nd degree polynomial equation (b) Polynomial expression
(c) Radical equation (d) All of these
56. Degree of Quadratic equation is
(a) 0 (b) 1 (c) 2 (d) None
57. Graph of quadratic equation is
(a) Straight line (b) Circle (c) Square (d) Parabola
58. Basic techniques for solving quadratic equations is/are
(a) 1 (b) 2 (c) 3 (d) 4
59. To solve $ax^2 + bx + c = 0$ where $a, b, c \in R$ & $a \neq 0$, we can use
(a) Factorization (b) Completing square
(c) Quadratic formula (d) All of these

60. The equation of the form $(x+a)(x+b)(x+c)(x+d)=k$ Where $a+b=c+d$, can be converted into
 (a) Reciprocal equation (b) Quadratic equation
 (c) Exponential equation (d) All of these
61. For any $n \in \mathbb{Z}$, ω^n is equivalent to one of
 (a) 1, ω , ω^2 (b) ω , ω^2 (c) 1, ω (d) 1, ω^2
62. $\omega^{28} + \omega^{29} + 1 =$
 (a) 0 (b) 1 (c) -1 (d) ω
63. Four forth roots of unity are
 (a) $\pm 1, \pm i$ (b) 0, ω, ω^2 (c) 1, $\frac{-1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}, 0$ (d) Non
64. Synthetic division is a process of
 (a) addition (b) multiplication (c) subtraction (d) division
65. $x^2 + x - 6 = 0$ has roots:
 (a) Real (b) Equal (c) Complex (d) Trivial
66. Roots of equation $x^2 + 2x + 3 = 0$ are:
 (a) real (b) equal (c) rational (d) imaginary
67. If the roots $px^2 + qx + 1 = 0$ are equal, then:
 (a) $q^2 + 4p = 0$ (b) $p^2 + 4q = 0$ (c) $q^2 - 4p = 0$ (d) $p^2 - 4q = 0$
68. A quadratic equation $Ax^2 + Bx + C = 0$ becomes linear equation if:
 (a) $C = 0$ (b) $A = 0$ (c) $B = 0$ (d) $A = B = C$

Chapter - 5

Multiple Choice Questions

(Encircle the correct answer choice)

- An open sentence formed by using sign of '=' is called a /an
 (a) equation (b) formula (c) Rational fraction (d) Theorem
- If an equation is true for all values of the variable, then it is called
 (a) a conditional equation (b) an identity
 (c) proper rational fraction (d) All of these
- If an equation is true only for particular values of the variable, then it is
 (a) a conditional equation (b) an identity
 (c) proper rational fraction (d) a formula
- $(x+3)(x+4) = x^2 + 7x + 12$ is a / an
 (a) conditional equation (b) identity
 (c) proper fraction (d) Linear factors
- $\sin^2 \theta + \cos^2 \theta$ is a / an
 (a) conditional equation (b) identity
 (c) proper fraction (d) Theorem

6. To express a single rational function as a sum of two or more single rational functions is called-
 - (a) partial fractions
 - (b) partial fraction resolution
 - (c) proper fraction
 - (d) Improper fraction
7. When a single rational fraction is expressed as a sum of two or more single rational fractions, then each single fraction is called
 - (a) partial fractions
 - (b) partial fraction resolution
 - (c) proper fraction
 - (d) Improper fraction
8. The value of a , when $(a+b)^2 = a^2 + 2ab + b^2$ is an identity
 - (a) an integer only
 - (b) any real number
 - (c) only positive number
 - (d) cannot be determined
9. If $\left| \begin{matrix} 7x & 3x \\ 2x^2 & p \end{matrix} \right| = 7xp - 6x^3$ is a/an
 - (a) equation
 - (b) identity
 - (c) determinant
 - (d) Non
10. The quotient of two polynomials $\frac{p(x)}{q(x)}$, $q(x) \neq 0$ is called
 - (a) Rational fraction
 - (b) An irrational fraction
 - (c) Proper fraction
 - (d) Partial fraction
11. A fraction $\frac{p(x)}{q(x)}$ is a proper fraction if
 - (a) degree of $p(x) <$ degree of $q(x)$
 - (b) degree of $p(x) =$ degree of $q(x)$
 - (c) degree of $p(x) >$ degree of $p(x)$
 - (d) degree of $p(x) \geq$ degree of $q(x)$
12. A fraction $\frac{p(x)}{q(x)}$ is an improper rational fraction if
 - (a) degree of $p(x) <$ degree of $q(x)$
 - (b) degree of $p(x) =$ degree of $q(x)$
 - (c) degree of $p(x) \leq$ degree of $p(x)$
 - (d) degree of $p(x) \geq$ degree of $q(x)$
13. A mixed form of fraction is
 - (a) an integer + improper fraction
 - (b) a polynomial + improper fraction
 - (c) a polynomial + proper fraction
 - (d) a polynomial + rational fraction
14. When a rational fraction is separated into partial fractions, then Result is always
 - (a) a conditional equations
 - (b) an identity
 - (c) a partial fraction
 - (d) an improper fraction
15. The partial fractions of $\frac{x^2 - 10x + 13}{(x-1)(x^2 - 5x + 6)}$ are of the form
 - (a) $\frac{A}{x-1} + \frac{Bx+C}{x-3} + \frac{Dx+E}{x-2}$
 - (b) $\frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$
 - (c) $\frac{Ax+B}{x-1} + \frac{C}{x^2-5x+6}$
 - (d) None of these

16. $\frac{x^2 - 5x + 7}{(x-1)(x^2 - 1)} = \frac{A}{x-1} + \dots$

- (a) $\frac{B}{x+1}$ (b) $\frac{B}{(x-1)^2} + \frac{C}{x+1}$ (c) $\frac{B}{x-1} + \frac{C}{x+1}$ (d) $\frac{Dx+E}{x^2-1}$

17. The number of partial fraction of $\frac{x^3}{x(x+1)(x^2-1)}$ are

- (a) 2 (b) 3 (c) 4 (d) none of these

18. The number of partial fraction of $\frac{x^5}{x(x+1)(x^2-4)}$ are

- (a) 3 (b) 4 (c) 5 (d) 6

19. The number of partial fraction of $\frac{x^4}{x^3-1}$ are

- (a) 1 (b) 2 (c) 3 (d) none of these

20. If $\frac{7x+25}{(x+3)(x+4)} = \frac{A}{x+3} + \frac{B}{x+4}$, then B is equal to

- (a) 3 (b) -3 (c) 4 (d) -4

21. If $\frac{x^2 - 10x + 13}{(x-1)(x^2 - 5x + 6)} = \frac{A}{x-2} + \frac{B}{x-3} + \frac{C}{x-1}$, then C is equal to

- (a) 2 (b) 3 (c) 4 (d) -4

22. If $\frac{2x^2 + x^2 - x - 3}{x(2x+3)(x-1)} = \frac{A}{x} + \frac{8}{2x+3} + \frac{C}{x-1}$, then A is equal to

- (a) 1 (b) 2 (c) x (d) none of these

23. Partial fractions of $\frac{x^2+1}{(x-1)(x+1)}$ are of the form

- (a) $\frac{Ax+B}{x^2-1}$ (b) $\frac{A}{x-1} + \frac{B}{x+1}$ (c) $1 + \frac{A}{x-1} + \frac{B}{x+1}$ (d) $1 + \frac{Ax+B}{x^2-1}$

24. If $\frac{1}{(x+1)^2(x^2-1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{(x+1)^3}$, then A =

- (a) $\frac{1}{8}$ (b) $-\frac{1}{2}$ (c) $-\frac{1}{8}$ (d) $-\frac{1}{4}$

25. A quadratic factor which can not be written as a product of linear factors with real coefficients is called

- (a) an irreducible factor (b) reducible factor
(c) an irrational factor (d) an improper factor

26. Which is a reducible factor

- (a) $x^3 - 6x^2 + 8x$ (b) $x^2 + 16x$ (c) $x^2 + 5x - 6$ (d) all of these

27. Partial fraction of $\frac{1}{x^2 - 1} =$
- (a) $\frac{1}{2(x-1)} + \frac{1}{2(x+1)}$ (b) $\frac{1}{2(x-1)} - \frac{1}{2(x+1)}$
- (c) $-\frac{1}{2(x-1)} + \frac{1}{2(x+1)}$ (d) $-\frac{1}{2(x-1)} - \frac{1}{2(x+1)}$
28. Partial fraction of $\frac{x^2 + 1}{x^3 + 1}$ will be of the form
- (a) $\frac{A}{x-1} - \frac{B}{x^2 - x + 1}$ (b) $\frac{A}{x+1} - \frac{B}{x^2 - x + 1}$
- (c) $\frac{A}{x+1} + \frac{Bx+c}{x^2 - x + 1}$ (d) $\frac{A}{x+1} - \frac{Bx+c}{x^2 - x - 1}$
29. Number of partial fractions of the fraction $\frac{1}{x(x-1)^3}$ are:
- (a) 1 (b) 2 (c) 3 (d) 4
30. Conditional equation $2x + 3 = 0$ holds when x is equal to:
- (a) $-\frac{3}{2}$ (b) $\frac{3}{2}$ (c) $\frac{1}{3}$ (d) 1
31. The quotient of two polynomials $\frac{P(x)}{Q(x)}$, $Q(x) \neq 0$ with no common factor is called:
- (a) algebraic relation (b) rational fraction
- (c) partial fraction (d) polynomial
32. The partial fractions of $\frac{1}{(x+1)(x-1)}$ are:
- (a) $\frac{1}{2(x-1)} - \frac{1}{2(x+1)}$ (b) $\frac{1}{2(x+1)} - \frac{1}{2(x-1)}$
- (c) $\frac{1}{2(x-1)} + \frac{1}{2(x+1)}$ (d) $-\frac{1}{2(x+1)} - \frac{1}{2(x-1)}$

Chapter - 6

Multiple Choice Questions

(Encircle the correct answer choice)

1. An arrangement of numbers according to some definite rule is called
- (a) Sequence (b) Combination (c) Series (d) Permutation

2. A sequence is also known as
 - (a) Real sequence
 - (b) Progression
 - (c) Arrangement
 - (d) Complex sequence
3. A sequence is a function whose domain is set of
 - (a) Integers ($\mathbb{Z}$)
 - (b) Rational numbers ($\mathbb{Q}$)
 - (c) Natural numbers
 - (d) real number
4. A sequence whose range is $\mathbb{R}$ i.e. set of real numbers, is called
 - (a) Real sequence
 - (b) Imaginary sequence
 - (c) Natural sequence
 - (d) Complex sequence
5. If $a_n = \{n + (-1)^n\}$, then $a_{10} =$
 - (a) 10
 - (b) 9
 - (c) 11
 - (d) none of these
6. The last term of an infinite sequence
 - (a) is n^{th} term
 - (b) is a_n
 - (c) is general term
 - (d) does not exist
7. The next term of the sequence 1, 2, 12, 40 ... is
 - (a) 112
 - (b) 120
 - (c) 124
 - (d) none of these
8. If $a_n - a_{n-1} = n + 1$ and $a_4 = 14$ then $a_5 =$
 - (a) 3
 - (b) 5
 - (c) 14
 - (d) 20
9. If $a_n = n a_{n-1}$, $a_1 = 1$ then $a_4 = ?$
 - (a) 6
 - (b) 24
 - (c) 110
 - (d) 660
10. A sequence $\{a_n\}$ in which $a_n - a_{n-1}$ is the same number for all $n \in \mathbb{N}$, $n > 1$, is called
 - (a) A.P
 - (b) G.P
 - (c) H.P
 - (d) none of these
11. If $\{a_n\}$ is an Arithmetic sequence then common difference is
 - (a) $a_{n+1} - a_{n-1}$
 - (b) $a_{n+1} - a_n$
 - (c) $a_n - a_{n+1}$
 - (d) $a_{n-1} - a_{n+1}$, $n \in \mathbb{N}$, $n > 1$
12. The general term of an A.P is
 - (a) $a_n = a + (n-1)d$
 - (b) $a_n = a - (n-1)d$
 - (c) $a_n = a + (n+1)d$
 - (d) $a_n = a - (n+1)d$
13. If $a_n = 5 - 3n + 2n^2$, then $a_{2n} =$
 - (a) $5 - 6n + 2n^2$
 - (b) $5 - 6n + 4n^2$
 - (c) $5 + 6n + 4n^2$
 - (d) $5 - 6n + 8n^2$
14. If $a_{n-2} = 3n - 11$, then $a_n =$
 - (a) $3n + 5$
 - (b) $3n - 5$
 - (c) $3n - 9$
 - (d) $3n - 13$
15. If n^{th} term of an A.P is $3n - 1$ then 10^{th} term is
 - (a) 9
 - (b) 29
 - (c) 12
 - (d) cannot be determined
16. n^{th} term of the series $\left(\frac{1}{3}\right)^2 + 1 + \left(\frac{5}{3}\right)^2 + \left(\frac{7}{3}\right)^2 + \dots$
 - (a) $\left(\frac{2n-1}{3}\right)^2$
 - (b) $\left(\frac{2n+1}{3}\right)^2$
 - (c) $\left(\frac{2n}{3}\right)^2$
 - (d) cannot be determined
17. If a_{n-1} , a_n , a_{n+1} are in A.P, then a_n is called
 - (a) A.M
 - (b) G.M
 - (c) H.M
 - (d) Mid point

18. Arithmetic mean between c and d is

- (a) $\frac{c+d}{2}$ (b) $\frac{c+d}{2cd}$ (c) $\frac{2cd}{c+d}$ (d) $\frac{2}{c+d}$

19. If a_{n-1}, a_n, a_{n+1} are in A.P then $a_n =$

- (a) $\frac{a_{n-1} + a_{n+1}}{2}$ (b) $\frac{a_{n+1} - a_{n-1}}{2}$ (c) $a_{n+1} - a_{n-1}$ (d) $\frac{a_{n-1} - a_{n+1}}{2}$

20. The Arithmetic mean between $\sqrt{2}$ and $3\sqrt{2}$ is

- (a) $4\sqrt{2}$ (b) $\frac{4}{\sqrt{2}}$ (c) $\sqrt{2}$ (d) none of these

21. The sum of terms of a sequence is called

- (a) Partial sum (b) Series (c) Finite sum (d) none of these

22. Forth partial sum of the sequence $\{n^2\}$ is

- (a) 16 (b) $1 + 4 + 9 + 16$ (c) 8 (d) $1 + 2 + 3 + 4$

23. Sum of n term of an Arithmetic series S_n is equal to

- (a) $\frac{n}{2} [2a + (n-1)d]$ (b) $\frac{n}{2} [a + (n-1)d]$
(c) $\frac{n}{2} [2a + (n+1)d]$ (d) $\frac{n}{2} (2a + l)$

24. Sum of n term of an Arithmetic series in S_n is equal to

- (a) $\frac{n}{2} (a_1 + a_n)$ (b) $\frac{n}{2} (a_1 - 1)$ (c) $\frac{a+a_n}{2}$ (d) $n(a_1 + a_n)$

25. For any G.P the common ratio r is equal to

- (a) $\frac{a_n}{a_{n+1}}$ (b) $\frac{a_{n-1}}{a_n}$ (c) $\frac{a_n}{a_{n-1}}$ (d) $a_{n+1} - a_n$ for $n \in \mathbb{N}, n > 1$

26. No term of a G.P is

- (a) 0 (b) 1 (c) negative (d) imaginary number

27. The general term of a G.P is

- (a) $a_n = ar^{n-1}$ (b) $a_n = ar^n$ (c) $a_n = ar^{n+1}$ (d) $a_n = \frac{a}{r^n - 1}$

28. If a, G, b are in G.P, then

- (a) $G = ab$ (b) $G = \pm \sqrt{ab}$ (c) $G = \frac{a+b}{2}$ (d) $G = \frac{2ab}{a+b}$

29. If a, G, b are in G.P, then G is called

- (a) common ratio (b) Geometric mean
(c) centre (d) Geometric series

30. If $G_1, G_2, G_3, \dots, G_n$ be Geometric means between a and b , then $G =$

- (a) $\sqrt[n]{G_1 G_2 \dots G_n}$ (b) $(G_1 G_2 \dots G_n)^{\frac{1}{n}}$
(c) $\frac{G_1 + G_2 + \dots + G_n}{n}$ (d) $\frac{1}{n} (G_1, G_2, \dots, G_n)$

31. Sum of n term of a geometric series S_n is equal to

- (a) $\frac{a(1-r^n)}{1-r}$ (b) $\frac{a(1-r^{n-1})}{1-r}$ (c) $\frac{a(r^n-1)}{1-r}$ (d) ar^{n-1} , for $r \neq 1$

32. The sum of infinite geometric series is valid if

- (a) $|r| > 1$ (b) $|r| = 1$ (c) $|r| \geq 1$ (d) $|r| < 1$

33. For the series $1 + 5 + 25 + 125 + \dots + \infty$, the sum is

- (a) -4 (b) 4 (c) $\frac{1-5^n}{-4}$ (d) not defined

34. An infinite geometric series is convergent if

- (a) $|r| > 1$ (b) $|r| = 1$ (c) $|r| \geq 1$ (d) $|r| < 1$

35. An infinite geometric series is Divergent if

- (a) $|r| < 1$ (b) $|r| \neq 1$ (c) $r = 0$ (d) $|r| > 1$

36. If sum of a series is defined, then it is called

- (a) Convergent series (b) Divergent series
(c) finite series (d) Geometric series

37. If sum of a series is not defined, then it is called

- (a) Convergent series (b) Divergent series
(c) finite series (d) Infinite series

38. If the series $\frac{x}{2} + \frac{x^2}{4} + \frac{x^3}{8} + \dots$ is convergent, then

- (a) $|x| \leq 2$ (b) $|x| \leq 1$ (c) $0 < x < 2$ (d) $|x| \geq 2$

39. If the series $\frac{2}{3}x + \frac{4}{9}x^2 + \frac{8}{27}x^3 + \dots$ is Divergent, then

- (a) $|\frac{2}{3}x| < 1$ (b) $\frac{2}{3}|x| < 1$ (c) $|x| \geq 1$ (d) $|\frac{2}{3}x| \geq 1$

40. The interval in which series $1 + 2x + 4x^2 + 8x^3 + \dots$ is convergent is

- (a) $-2 < x < 2$ (b) $-\frac{1}{2} < x < \frac{1}{2}$ (c) $|2x| > 1$ (d) $|x| < 1$

41. If the reciprocals of the terms of a sequence form an A.P, then it is

- (a) Harmonic sequence (b) Arithmetic sequence
(c) Reciprocal sequence (d) series

42. The n^{th} term of $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \dots$ is

- (a) $\frac{1}{3n-1}$ (b) $3n-1$ (c) $2n+1$ (d) $\frac{1}{3n+1}$

43. General term of an H.P is

- (a) $a_n = \frac{1}{a + (n+1)d}$ (b) $a_n = \frac{1}{a + (n-1)d}$
(c) $a_n = \frac{1}{a + nd}$ (d) $a_n = a + (n-1)d$

44. Harmonic mean between 2 and 8 is
 (a) 5 (b) $\frac{16}{5}$ (c) ± 4 (d) $\frac{5}{16}$
45. If A, G, and H are Arithmetic, Geometric and Harmonic means between two positive number, then
 (a) $G^2 = A H$ (b) A, G, H are in G.P
 (c) $A > G > H$ (d) all of these
46. If A, G, and H are Arithmetic, Geometric and Harmonic means between two negative number, then
 (a) $G^2 = A H$ (b) A, G, H are in G.P
 (c) $A < G < H$ (d) all of these
47. If a and b are two negative number, then
 (a) $A < G < H$ (b) $A > G > H$
 (c) $A = G = H$ (d) $A \geq G \geq H$
48. If a and b are two positive number, then
 (a) $A < G < H$ (b) $A > G > H$
 (c) $A = G = H$ (d) $A \leq G \leq H$
49. If a and b have opposite signs then Geometric mean is
 (a) an imaginary number (b) non zero real number
 (c) Real number (d) Negative
50. If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is A.M between a & b , then n is equal to
 (a) 0 (b) -1 (c) 1 (d) $\frac{1}{2}$
51. If $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is G.M between a & b , then n is equal to
 (a) 0 (b) -1 (c) 1 (d) $\frac{1}{2}$
52. If $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ is H.M between a & b , then n is equal to
 (a) 0 (b) -1 (c) 1 (d) $\frac{1}{2}$
53. If $a, ar^2, ar^4, \dots$ form a G. P then $\frac{1}{a}, \frac{1}{ar^2}, \frac{1}{ar^4}, \dots$ is
 (a) an A. P (b) a G. P
 (c) an H. P (d) a reciprocal sequence
54. $\sum n$ is equal to
 (a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{n^2(n+1)^2}{2}$ (d) n^2
55. $\sum n^2$ is equal to
 (a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{n^2(n+1)^2}{2}$ (d) n^2

56. $\sum n^3$ is equal to
 (a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{n^2(n+1)^2}{4}$ (d) $\frac{n(n+1)^2}{2}$
57. If $S_n = (n+1)^2$, then S_{2n} is equal to
 (a) $2n+1$ (b) $4n^2+4n+1$
 (c) $(2n-1)^2$ (d) cannot be determined
58. The sum of n A.Ms between a & b is equal to
 (a) $n\left(\frac{a+b}{2}\right)$ (b) $n(a+b)$
 (c) $n[a+(n-1)d]$ (d) $a+(n-1)d$
59. The sum of 5 A.Ms between 2 & 8 is
 (a) 25 (b) 50 (c) 40 (d) 10
60. If both x & y are +ve distinct real numbers, then the G.M. between is
 (a) less than their A.M. (b) Equal to A. M.
 (c) Greater than A.M. (d) None of these
61. The numbers $a-d, a, a+d$ are in
 (a) arithmetic progression (b) geometric progression
 (c) harmonic progression (d) harmonic series
62. If $|r| < 1$, then, $S_n =$
 (a) $\frac{a_1(1-r^n)}{1-r}$ (b) $\frac{a_1(r^n-1)}{1-r}$ (c) $\frac{a_1(r^n-1)}{r-1}$ (d) $\frac{a_1(1-r^n)}{r-1}$
63. With usual notations, AH equals
 (a) A^2 (b) H^2 (c) G^2 (d) $-G^2$
64. If $a_{n-1} = 2n+1$, then a_n is equal to
 (a) $2n+3$ (b) $2n-3$ (c) $2n-1$ (d) $2n+1$
65. With usual notation n th term of A.P is
 (a) $a_n = a_1 + (n+1)d$ (b) $a_n = a_1 - (n-1)d$
 (c) $a_n = a_1 + a_{n+1}$ (d) $a_n = a_1 + (n-1)d$
66. G. M between -2 and 8 is:
 (a) $4i$ or $-4i$ (b) 4 or -4 (c) 16 or -16 (d) 3 or -5
67. H. M between -2 and 8 equals:
 (a) $-\frac{3}{16}$ (b) $-\frac{16}{3}$ (c) $-\frac{16}{5}$ (d) $-\frac{5}{16}$
68. n^{th} term of A.P is:
 (a) $a_1 + nd$ (b) $a_1 + (n-1)d$ (c) $na_1 + d$ (d) $\frac{a_1}{n} + d$
69. Fifth term of $\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \dots$ is
 (a) $1/9$ (b) 9 (c) $1/11$ (d) 11

70. $\frac{1}{2}, \frac{1}{7}, \frac{1}{12}, \dots$ is:
 (a) An A.P (b) G.P (c) H.P (d) Harmonic series
71. If $G_1, G_2, \dots, G_n$ are n geometric means between a and b , then $(G_1 \cdot G_2 \cdot \dots \cdot G_n)^{1/n}$ is.
 (a) $\frac{a+b}{2}$ (b) $\frac{2ab}{a+b}$ (c) $\sqrt{ab}$ (d) $\frac{a+b}{2ab}$
72. Harmonic mean between two numbers ' a ' and ' b ' is:
 (a) $\frac{a+b}{2}$ (b) $\pm \sqrt{ab}$ (c) $\frac{2ab}{a+b}$ (d) $\frac{a+b}{2ab}$
73. General term of a sequence is $(-1)^n n^2$. Its 4th term is:
 (a) -4 (b) -16 (c) 16 (d) 4

Chapter - 7

Multiple Choice Questions

(Encircle the correct answer choice)

- The factorial notation was introduced by
 (a) Christian kramp (b) Newton (c) Candy (d) Boyal
- $n! = n(n-1)(n-2) \dots 3.2.1$ is defined only when n is
 (a) positive integer (b) an integer
 (c) Real number (d) whole number
- $0!$ is equal to
 (a) 0 (b) 1 (c) -1 (d) not defined
- $(-1)!$ is equal to
 (a) 0 (b) 1 (c) -1 (d) not defined
- The factorial form of $12.11.10.$ is
 (a) $\frac{12!}{9!}$ (b) $12!$ (c) $(\frac{12}{9})!$ (d) $(12!)(9!)$
- The factorial form of $n(n-1)(n-2) \dots (n-r+1)$ is
 (a) $\frac{n!}{(n-r)!}$ (b) $\frac{n!}{n-r!}$ (c) $\frac{n!}{(n-r-1)!}$ (d) $\frac{n!}{(n-r+1)!}$
- The factorial form of $6.5.4$ is
 (a) $(\frac{6}{3})!$ (b) $6!$ (c) $5!$ (d) None of these
- If an event A can occurs in p ways B can occur in q ways, then number of way that both events can occur is
 (a) $p+q$ (b) $p \cdot q$ (c) $(pq)!$ (d) $(p+q)!$
- An arrangement of n objects according to some definite order is called
 (a) Combination (b) permutation
 (c) factorial (d) ordered arrangement

10. An arrangement of n objects without any order is called
 (a) Combination (b) permutation
 (c) factorial (d) ordered arrangement
11. An arrangement of n objects taking r out of them at a time without any order is
 (a) nC_r (b) nP_r (c) $(n+r)!$ (d) $(nr)!$
12. An arrangement of n objects taking r out of them at a time with some definite order is
 (a) nC_r (b) nP_r (c) $(n+r)!$ (d) $(nr)!$
13. $8 \cdot 7 \cdot 6$ is equal to
 (a) 8P_3 (b) 8C_3 (c) 8P_5 (d) 8C_5
14. In a permutation nP_r or $P(n, r)$, it is always true that
 (a) $n \geq r$ (b) $n < r$ (c) $n \leq r$ (d) $n < 0, r < 0$
15. Different signals of 5 flags of different colours, using 3 at a time is
 (a) 60 (b) 5 (c) 120 (d) 10
16. If $r = n$, then nP_r is equal to
 (a) $r!$ (b) $(n-r)!$ (c) 1 (d) 0
17. ${}^{10}P_7$ is equal to
 (a) 10 (b) 720 (c) 120 (d) non of these
18. If there are p like object of one kind and q like object of 2nd kind out of n objects, then different permutation are
 (a) $\frac{n!}{(n-p)!(n-q)!}$ (b) $\frac{n!}{(pq)!}$ (c) $\frac{n!}{p!q!}$ (d) $n! - 1$
19. Different circular permutations of n objects are
 (a) $n!$ (b) $(n-1)!$ (c) $(n+1)!$ (d) $n! - 1$
20. The number of ways that a necklace of n beads of different colours be made is
 (a) $n!$ (b) $\frac{n!}{2}$ (c) $\frac{n! - 1}{2}$ (d) $\frac{(n-1)!}{2}$
21. The numbers of permutations of the word PANAMA are
 (a) 120 (b) 20 (c) 10 (d) 60
22. The numbers of permutation of the word PANAMA when each word starts with P is
 (a) 120 (b) 20 (c) 10 (d) 60
23. 5 Persons can be seated at a round table in ways
 (a) 120 (b) 24 (c) 720 (d) 12
24. nP_r is equal to
 (a) $\frac{n!}{r!}$ (b) $\frac{n!}{r!(n-r)!}$ (c) $\frac{n!}{(n-r)!}$ (d) $\frac{n!}{r!(n+r)!}$
25. nC_r is equal to
 (a) $\frac{n!}{r!}$ (b) $\frac{n!}{r!(n-r)!}$ (c) $\frac{n!}{(n-r)!}$ (d) $\frac{n!}{r!(n+r)!}$

26. Complementary combination is
 (a) ${}^nC_r = {}^nC_{r-1}$ (b) ${}^nC_r = {}^nC_{n-r}$ (c) ${}^nC_{r+1} = {}^nC_{r-1}$ (d) ${}^nC_r = {}^nP_r$
27. If ${}^nC_8 = {}^nC_{12}$, then n is equal to
 (a) 8 (b) 12 (c) 20 (d) 4
28. The number of Triangles of an n sided polygon is
 (a) nC_3 (b) nP_3 (c) ${}^nP_3 - n$ (d) ${}^nC_3 - n$
29. ${}^{n-1}C_r + {}^{n-1}C_{r-1} =$
 (a) ${}^{n-1}C_r$ (b) nC_r (c) ${}^{n-1}C_{r-1}$ (d) ${}^nC_{r-1}$
30. ${}^nC_7 + {}^nC_8 =$
 (a) ${}^{n+1}C_7$ (b) ${}^{n+1}C_8$ (c) ${}^{n+1}C_9$ (d) nC_9
31. The number of Diagonals of a 5 sided polygon is
 (a) 5 (b) 20 (c) 15 (d) 10
32. The number of Triangles of a 5 sided Polygon is
 (a) 5 (b) 10 (c) 15 (d) 20
33. A hockey 11 out of 15 players be selected, different teams if a particular players must be selected is
 (a) ${}^{15}C_{11}$ (b) ${}^{15}P_{11}$ (c) ${}^{14}C_{10}$ (d) ${}^{14}C_{10}$
34. The set of all possible outcomes of an experiment is
 (a) Sample space (b) Event
 (c) Simple Event (d) Random Experiment
35. Any particular outcome of an experiment is called
 (a) Sample space (b) an Event
 (c) a Trial (d) Random Variable
36. A fair coin is tossed, the probability of getting a head or tail is
 (a) 1 (b) 0 (c) $\frac{1}{2}$ (d) $\frac{1}{4}$
37. For two events A and B if $A \cap B = \phi$, then events A and B are called
 (a) mutually exclusive (b) not mutually exclusive
 (c) Overlapping (d) dependent events
38. If A and B are mutually exclusive (Disjoint) events, then $n(A \cap B)$ is
 (a) 0 (b) 1 (c) between 0 and 1 (d) not defined
39. If two events A and B have equal chance of occurrence, then the events are
 (a) Equally likely (b) Not equally likely
 (c) Dependent (d) not mutually exclusive
40. If E be an event of a sample space S, then
 (a) $P(E) = \frac{n(E)}{n(S)}$ (b) $0 < P(E) \leq 1$ (c) $P(E) > 1$ (d) all of these
41. If E be an event of a sample space S, then
 (a) $P(E) = \frac{n(S)}{n(E)}$ (b) $0 \leq P(E) \leq 1$
 (c) $0 < P(E) < 1$ (d) all of these

42. If an event always occurs, then it is called
(a) Null event (b) possible event
(c) certain event (d) independent event
43. If E is a certain event, then
(a) $P(E) = 0$ (b) $P(E) = 1$ (c) $0 < P(E) < 1$ (d) $P(E) > 1$
44. If E is an impossible event, then
(a) $P(E) = 0$ (b) $P(E) = 1$ (c) $P(E) \neq 0$ (d) $0 < P(E) < 1$
45. Non occurrence of an event E is denoted by
(a) $\sim E$ (b) $\bar{E}$ (c) E^c (d) all of these
46. If E be an event of a sample space S , then
(a) $P(E) = 1 + P(\bar{E})$ (b) $P(\bar{E}) = 1 + P(E)$
(c) $P(E) = 1 - P(\bar{E})$ (d) $P(\bar{E}) = 1 - P(E)$
47. Let $S = \{1, 2, 3, \dots, 10\}$ the probability that a number is divisible by 4 is
(a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{1}{10}$ (d) $\frac{1}{2}$
48. There are 5 green and 3 red balls in a box. One ball is taken, the probability that ball is green or red is
(a) $\frac{3}{8}$ (b) 1 (c) $\frac{15}{8}$ (d) $\frac{15}{64}$
49. There are 5 green and 3 red balls in a box the one ball taken is probability of getting a black ball is.
(a) 0 (b) 1 (c) $\frac{15}{8}$ (d) $\frac{15}{64}$
50. Three dice are rolled simultaneously, then $n(S)$ is equal to
(a) 36 (b) 18 (c) 216 (d) 6
51. A coin is tossed 5 times, then $n(S)$ is equal to
(a) 32 (b) 25 (c) 10 (d) 20
52. A bag contain 40 balls out of which 15 are black, then probability of a ball not black is
(a) $\frac{3}{8}$ (b) $\frac{5}{8}$ (c) $\frac{15}{8}$ (d) $\frac{15}{64}$
53. Two teams A and B are playing a match, the probability that team A dose not loose is
(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) 1 (d) 0
54. If $P(E) = \frac{7}{12}$, $n(S) = 8400$, $n(E)$ is equal to
(a) 108 (b) 4900 (c) 144 (d) 14400
55. A die is rolled, the probability of getting 3 or 5 is
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{15}{36}$ (d) $\frac{1}{36}$

56. A die is rolled, the probability of getting 3 or an even number is
(a) $\frac{1}{12}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) non of these
57. A coin is tossed 4 times, then probability that at least one head appears in 4 tosses is
(a) $\frac{1}{16}$ (b) $\frac{15}{16}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$
58. If A and B are disjoint event, then $P(A \cup B)$ is equal to
(a) $P(A) + P(B)$ (b) $P(A) \cdot P(B)$
(c) $P(A) + P(B) - P(A \cap B)$ (d) $P(A \cap B)$
59. If A and B are over lapping event, then $P(A \cup B)$ is equal to
(a) $P(A) + P(B)$ (b) $P(A) \cdot P(B)$
(c) $P(A) + P(B) - P(A \cap B)$ (d) $P(A \cap B)$
60. If $S = \{1, 2, \dots, 10\}$, $A = \{1, 3, 5\}$ $B = \{2, 4, 6\}$ then $P(A \cup B)$ is equal to
(a) $\frac{3}{5}$ (b) $\frac{2}{5}$ (c) $\frac{9}{100}$ (d) 0
61. If two event do not effect the occurrence or non occurrence of each other then, these are called
(a) Independent events (b) Dependent events
(c) Equal events (d) Different events
62. If two event effect the occurrence or non occurrence of each other, then these are called
(a) Independent events (b) Dependent events
(c) Equal events (d) Different events
63. If A, B and C are Independent event, then $P(A \cap B \cap C)$ is equal to
(a) $P(A) + P(B) + P(C)$ (b) $P(A) \cdot P(B) \cdot P(C)$
(c) $P(A \cup B \cup C)$ (d) none
64. If A, B and C are disjoint events then $P(A \cup B \cup C)$ is equal to
(a) $P(A) + P(B) + P(C)$ (b) $P(A) \cdot P(B) \cdot P(C)$
(c) $P(A \cap B \cap C)$ (d) none
65. If $P(A) = \frac{5}{7}$, $P(B) = \frac{7}{9}$, $P(A \cap B)$ is equal to
(a) $\frac{5}{9}$ (b) $\frac{3}{4}$ (c) $\frac{94}{63}$ (d) None of these
66. If A and B are two independent events, $P(A \cap B) = \frac{1}{169}$, $P(A) = \frac{1}{13}$, $P(B) =$
(a) $\frac{1}{13}$ (b) $\frac{1}{2097}$ (c) 13 (d) $\frac{12}{169}$
67. The number of ways for sitting 4 persons in a train on a straight sofa is
(a) 24 (b) 6 (c) 4 (d) None of these
68. Four persons want to sit in a circular sofa, the total ways are
(a) 24 (b) 6 (c) 4 (d) None of these

69. A card is drawn from a deck of 52 playing cards. The probability of card that it is an ace card is
 (a) $\frac{2}{13}$ (b) $\frac{4}{13}$ (c) $\frac{1}{13}$ (d) $\frac{17}{13}$
70. If ${}^nC_6 = {}^nC_{12}$, then n equals
 (a) 18 (b) 12 (c) 6 (d) 20
71. For independent events $P(A \cap B) =$
 (a) $P(A) + P(B)$ (b) $P(A) - P(B)$
 (c) $P(A) \cdot P(B)$ (d) $\frac{P(A)}{P(B)}$
72. If $\binom{n}{12} = \binom{n}{8}$, then the value of $n =$
 (a) 15 (b) 16 (c) 18 (d) 20
73. If A and B are disjoint event then $P(A \cup B) =$
 (a) $P(A) + P(B)$ (b) $P(A) + P(B) - P(A \cap B)$
 (c) $P(A) - P(B) + P(A \cap B)$ (d) $P(A) \cdot P(B) - P(A \cap B)$
74. With usual notation nP_n equals:
 (a) n (b) 0 (c) $0!$ (d) $n!$
75. If ${}^nC_6 = {}^nC_8$ then, n equals:
 (a) 20 (b) 24 (c) 14 (d) -14
76. Sample space for tossing a coin is:
 (a) {H} (b) {T} (c) {H, H} (d) {H, T}
77. Probability of non-occurrence of an event E is equal to:
 (a) $1 - P(E)$ (b) $P(E) + \frac{n(s)}{n(E)}$ (c) $\frac{n(s)}{n(E)}$ (d) $1 + P(E)$

Chapter - 8

Multiple Choice Questions

(Encircle the correct answer choice)

- The statement $4^n > 3^n + 4$ is true when
 (a) $n = 0$ (b) $n = 1$
 (c) $n \geq 2$ (d) n is any positive integer
- The statement $3^n < n!$ is true, when
 (a) $n = 2$ (b) $n = 4$ (c) $n = 6$ (d) $n > 6$
- The general term of the binomial expansion $(a + x)^n$ is
 (a) $\binom{n}{r} a^r x^r$ (b) $\binom{n}{r} a^{n-r} x^r$ (c) $\binom{n}{r} a^r r^{n-r}$ (d) $\binom{n}{r} (a x)^{n-r}$
- The number of terms in the expansion of $(a + b)^n$ are
 (a) n (b) $n + 1$ (c) 2^n (d) 2^{n-1}

5. In the expansion $(a + x)^n$, the sum of exponents of a and x is
 (a) n (b) $n - 1$ (c) $n + 1$ (d) $2n$
6. The $(r + 1)^{\text{th}}$ term in the expansion of $(a + x)^n$ is
 (a) $\binom{n}{r+1} a^{n-r} x^r$ (b) $\binom{n}{r} a^{n-r} x^r$
 (c) $\binom{n}{r} a^{n-r+1} x^{r+1}$ (d) $\binom{n}{r} a^{n-(r+1)} x^{r+1}$
7. In the expansion $(a + x)^n$ the exponent of 'a'
 (a) decreases from n to 0 (b) Increases from 0 to n
 (c) remains n every where (d) becomes n at the end
8. In the expansion $(a + x)^n$ the exponent of 'x'
 (a) decreases from n to 0 (b) Increases from 0 to n
 (c) remains n every where (d) becomes 0 at the end
9. Middle term/s in the expansion of $(a + b)^{11}$ is/ are
 (a) T_6 (b) T_5 & T_6 (c) T_6 & T_7 (d) T_5
10. Middle term/s in the expansion of $(a - 3x)^{14}$ is/ are
 (a) T_7 (b) T_8 (c) T_6 & T_7 (d) T_7 & T_8
11. 6th term of the expansion $(a + 2x)^{13}$ is
 (a) $\binom{13}{5} a^8 \cdot x^5$ (b) $\binom{13}{5} a^8 \cdot 2^5 \cdot x^5$ (c) $\binom{13}{8} a^5 \cdot x^8$ (d) $\binom{13}{8} a^5 \cdot 2^8 \cdot x^8$
12. 4th term from the end in the expansion of $(a + b)^9$ is
 (a) T_6 (b) T_4 (c) T_7 (d) non of these
13. The term independent of x in the expansion of $(a + 2x)^n$ is
 (a) first term (b) Middle term (c) last term (d) 2nd last term
14. The coefficient of the last term in the expansion of $(2 - x)^7$ is
 (a) 1 (b) -1 (c) 7 (d) -7
15. Sum of all binomial coefficients in the expansion of $(a + x)^n$ is
 (a) 2^n (b) 2^{n-1} (c) 2^{n+1} (d) $n + 1$
16. Sum of odd binomial coefficients in the expansion of $(a + x)^n$ is
 (a) 2^n (b) 2^{n-1} (c) 2^{n+1} (d) $n + 1$
17. Sum of even binomial coefficients in the expansion of $(a + x)^n$ is
 (a) 2^n (b) 2^{n-1} (c) 2^{n+1} (d) $n + 1$
18. $\binom{n+1}{0} + \binom{n+1}{1} + \binom{n+1}{2} + \dots + \binom{n+1}{n+1}$ is equal to
 (a) 2^n (b) 2^{n+1} (c) 2^{n-1} (d) cannot be determined
19. $\binom{2n}{0} + \binom{2n}{1} + \binom{2n}{2} + \dots + \binom{2n}{2n}$ is equal to
 (a) 2^n (b) 2^{2n} (c) 2^{2n-1} (d) 2^{2n+1}

20. If n is odd, the middle term/s in $(a+x)^n$ is/ are

- (a) $(\frac{n+1}{2})^{\text{th}}$ (b) $(\frac{n}{2}+1)^{\text{th}}$
 (c) $(\frac{n+1}{2})^{\text{th}}$ & $(\frac{n+3}{2})^{\text{th}}$ (d) $(\frac{n}{2}+1)^{\text{th}}$ & $(\frac{n}{2}+2)^{\text{th}}$

21. If n is even the middle terms in $(a+x)^n$ is

- (a) $(\frac{n+1}{2})^{\text{th}}$ (b) $(\frac{n}{2}+1)^{\text{th}}$
 (c) $(\frac{n+1}{2})^{\text{th}}$ & $(\frac{n+3}{2})^{\text{th}}$ (d) $(\frac{n}{2}+1)^{\text{th}}$ & $(\frac{n}{2}+2)^{\text{th}}$

22. Which term of $(x+2)^8$ is independent of x

- (a) First (b) Second (c) Middle (d) Last

23. The series $(1+x)^n$ is valid if

- (a) $x < 1$ (b) $-1 < x < 1$ (c) $x > 1$ (d) $x = 1$

24. $1+x+x^2+x^3+\dots$ is equal to

- (a) $(1+x)^{-1}$ (b) $(1-x)^{-1}$ (c) $(1+x)^{-2}$ (d) $(1-x)^{-2}$

25. $1-x+x^2-x^3+\dots$ is equal to

- (a) $(1+x)^{-1}$ (b) $(1-x)^{-1}$ (c) $(1+x)^{-2}$ (d) $(1-x)^{-2}$

26. When n is negative or fraction, then general term of $(1+x)^n$ is

- (a) $T_r = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} x^r$
 (b) $T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!} x^r$
 (c) $T_r = n(n-1)(n-2)\dots 3.2.1. x^r$ (d) $T_{r+1} = \binom{n}{r} x^r$

27. If $T_{r+1} = \binom{10}{r} (-2)^r (x)^{10-2r}$, The term independent of x is

- (a) 10th (b) 5th (c) 4th (d) 6th

28. The sum of exponents of a and b in every term of the expansion $(a+b)^n$ is

- (a) 1 (b) 0 (c) $2n$ (d) n

29. The expansion of $(1-2x)^{-2}$ is valid if

- (a) $|x| < 0$ (b) $|x| < \frac{1}{2}$ (c) $|x| < 2$ (d) $|x| < 1$

30. $n^2 > n+3$ is true for:

- (a) $n \geq 3$ (b) $n \geq 1$ (c) $n \geq 2$ (d) $n \geq -1$

31. If n is odd number, then middle term in expansion $(a+x)^n$ is:

- (a) $\frac{n+1}{2}$ (b) $\frac{n+3}{2}$ (c) $\frac{n-1}{2}$ (d) $\frac{n+1}{2}$ and $\frac{n+3}{2}$

32. The expansion $(1 - 4x)^{-2}$ is valid if:
 (a) $|x| < \frac{1}{4}$ (b) $|x| > \frac{1}{4}$ (c) $-1 < x < 1$ (d) $|x| < -1$
33. The middle term in the expansion of $(a + b)^n$ is $\left(\frac{n}{2} + 1\right)$; then n is:
 (a) odd (b) even (c) prime (d) none of these
34. Number of terms in the expansion of $(1 + x)^n$ is:
 (a) n (b) $n/2$ (c) $n - 1$ (d) $n + 1$
35. The number of terms in the expansion of $(a + b)^{20}$ is:
 (a) 18 (b) 20 (c) 21 (d) 19

Chapter - 9

Multiple Choice Questions

(Encircle the correct answer choice)

- Two rays with a common starting point form:
 (a) Triangle (b) Angle (c) Radian (d) Minute
- The common starting point of two rays is called:
 (a) Origin (b) Initial point
 (c) Vertex (d) All of these
- If the rotation of angle is counter clockwise, then angle is:
 (a) Negative (b) Positive
 (c) Non-negative (d) None of these
- If the initial ray $\overrightarrow{OA}$ rotates in anti-clockwise direction in such a way that it coincides with itself, the angle then formed is:
 (a) 180° (b) 270° (c) 300° (d) 360°
- One Rotation in anticlockwise direction is equal to
 (a) 180° (b) 270° (c) 360° (d) 90°
- Straight line angle is equal to
 (a) $\frac{1}{2}$ rotation (b) π radian (c) 180° (d) All of these
- One right angle is equal to is equal to
 (a) $\frac{\pi}{2}$ radian (b) 90° (c) $\frac{1}{4}$ rotation (d) All of these
- 1° is equal to
 (a) 30 minute (b) 60 minute (c) $\frac{1}{60}$ minute (d) $\frac{1}{2}$ minute
- 1° is equal to
 (a) $360''$ (b) $3600''$ (c) $\left(\frac{1}{360}\right)'$ (d) $60''$

10. 60th part of 1° is equal to
 (a) One Second (b) One minute (c) 1 Radian (d) π Radian
11. 60th part of 1' is equal to
 (a) 1' (b) 1'' (c) 60'' (d) 3600''
12. 3600th part of 1° is equal to
 (a) 1' (b) 1'' (c) 60'' (d) 3600''
13. Sexagesimal system is also called:
 (a) German system (b) English system
 (c) C.G.S system (d) S I system
14. 16°30' is equal to
 (a) 16.5° (b) $\frac{32^\circ}{2}$ (c) 16.05° (d) 16.2°
15. Conversion of 21.256° to D°m's" form is:
 (a) 21°, 25', 6" (b) 21°, 40', 27" (c) 21°, 15', 22" (d) 21°, 30', 2"
16. The angle subtended at the centre of the circle by an arc whose length is equal to the radius of the circle is called:
 (a) 1 Degree (b) 1' (c) 1 Radian (d) 1"
17. The system of angular measurement in which angle is measured in radian is called:
 (a) Sexagesimal system (b) Circular system
 (c) English system (d) Gradient system
18. Relation between the length of an arc of a circle and the circular measure of its central angle is:
 (a) $\ell = \frac{r}{\theta}$ (b) $\theta = \ell r$ (c) $\theta = \frac{\ell}{r}$ (d) $\ell = \frac{1}{2} r^2 \theta$
19. With usual notations, if $\ell = 6\text{cm}$, $r = 2\text{cm}$, then unit of θ is:
 (a) cm (b) cm² (c) No unit (d) cm³
20. 1° is equal to
 (a) $\left(\frac{\pi}{180}\right)^\circ$ (b) $\frac{180}{\pi}$ radian (c) $\left(\frac{180}{\pi}\right)^\circ$ (d) $\frac{\pi}{180}$ radian
21. 1° is equal to
 (a) 0.175 rad (b) 0.0175 rad (c) 1.75 rad (d) 0.00175 rad
22. 1 Radian is equal to
 (a) $\frac{\pi}{180}$ rad (b) $\frac{180}{\pi}$ rad (c) $\left(\frac{180}{\pi}\right)^\circ$ (d) $\left(\frac{\pi}{180}\right)^\circ$
23. 1 radian is equal to
 (a) 57.296° (b) 5.7296° (c) 175.27° (d) 17.5270
24. 3 radian
 (a) 171.888° (b) 120° (c) 300° (d) 270°

25. $105^\circ = \dots\dots\dots$ radian
(a) $\frac{7\pi}{12}$ (b) $\frac{2\pi}{3}$ (c) $\frac{5\pi}{12}$ (d) $\frac{5\pi}{6}$
26. $3'' = \dots$ radian
(a) $\frac{53\pi}{270}$ (b) $\frac{\pi}{216000}$ (c) $\frac{41\pi}{720}$ (d) $\frac{27721\pi}{32400}$
27. $\frac{\pi}{4}$ radian = $\dots$ deg
(a) 45° (b) 30° (c) 60° (d) 75°
28. Circular measure of angle between the hands of a watch at 4-o clock are
(a) 45° (b) 120° (c) $\frac{3\pi}{2}$ (d) 270°
29. If $\ell = 1.5$ cm & $r = 2.5$ c, then θ is equal to
(a) $\frac{3}{5}$ (b) $\frac{5}{3}$ (c) 3.75 (d) None
30. If $\theta = 45^\circ$, $r = 18$ mm, then $\ell =$
(a) $\frac{9}{2}\pi$ (b) $\frac{2}{9}\pi$ (c) 810 (d) 810 mm
31. Area of sector of circle of radius r is:
(a) $\frac{1}{2} r^2 \theta$ (b) $\frac{1}{2} r \theta^2$ (c) $\frac{1}{2} (r\theta)^2$ (d) $\frac{1}{2r^2\theta}$
32. Angles with same initial and terminal sides are called:
(a) Acute angles (b) Allied angles
(c) Conterminal angles (d) Quad rental angle
33. If angle θ is in degree, then the angle conterminal with θ is
(a) $\theta + 180^\circ k$, $k \in \mathbb{Z}$ (b) $\theta + 360^\circ k$, $k \in \mathbb{Z}$
(c) $\theta + 90^\circ k$, k (d) None of these
34. If angle θ is in radian then angle conterminal with θ is:
(a) $\theta + 2k\pi$, $k \in \mathbb{Z}$ (b) $\theta + k\pi$, $k \in \mathbb{Z}$
(c) $-\theta + 2k\pi$, $k \in \mathbb{Z}$ (d) $-(\theta + 2\pi)$
35. If the vertex lies at the origin of rectangular coordinate system and its initial side along the positive x -axis, then angle is called:
(a) Acute angle (b) Conterminal angle
(c) Angle in standard position (d) Quadrantal angle
36. An angle is in standard position, if its vertex lies
(a) at origin (b) at x -axis (c) at y -axis (d) in 1st Quadrant
37. If initial and the terminal side of an angle falls on x -axis or y -axis, then it is called:
(a) Coterminal angle (b) Quadrantal angle
(c) Allied angles (d) None of these

38. $0^\circ, 90^\circ, 180^\circ, 270^\circ$ & 360° are:
 (a) Coterminal angle (b) Quadrantal angle
 (c) Allied angles (d) None of these
39. $\sin^2\theta + \cos^2\theta$ is equal to
 (a) 0 (b) 1 (c) -1 (d) $\sec^2\theta$
40. $1 + \tan^2\theta$ is equal to
 (a) $\operatorname{cosec}^2\theta$ (b) $\sin^2\theta$ (c) $\sec^2\theta$ (d) $\cot^2\theta$
41. $\operatorname{cosec}^2\theta - \cot^2\theta$ is equal to
 (a) 0 (b) 1 (c) -1 (d) 2
42. If $\sin\theta < 0$ & $\cos\theta > 0$, then the terminal arm of angle lies in ... Quad.
 (a) I (b) II (c) III (d) IV
43. If $\cot\theta > 0$ & $\operatorname{cosec}\theta > 0$, then the terminal arm of angle lies in ... Quad.
 (a) I (b) II (c) III (d) IV
44. If $\tan\theta < 0$ & $\operatorname{cosec}\theta > 0$, then the terminal arm of angle lies in ... Quad.
 (a) I (b) II (c) III (d) IV
45. If $\sec\theta < 0$ & $\sin\theta < 0$, then the terminal arm of angle lies in ... Quad.
 (a) I (b) II (c) III (d) IV
46. In right angle Triangle, the measure of the side opposite to 30° is:
 (a) Half of Hypotenuse (b) Half of base
 (c) Double of base (d) None of these
47. The point (0, 1) lies on terminal side of the angle:
 (a) 0 (b) 90° (c) 180° (d) 270°
48. The point (-1, 0) lies on terminal side of angle:
 (a) 0 (b) 90° (c) 180° (d) 270°
49. The point (0, -1) lies on terminal side of angle:
 (a) 0 (b) 90° (c) 180° (d) 270°
50. $2 \sin 45^\circ + \frac{1}{2} \operatorname{cosec} 45^\circ =$
 (a) $\frac{\sqrt{2}}{3}$ (b) $\frac{3}{\sqrt{2}}$ (c) -1 (d) 1
51. Domain of $\sin\theta$ is
 (a) $\mathbb{R}$ (b) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$
 (c) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (d) None of these
52. Domain of $\cos\theta =$
 (a) $\mathbb{R}$ (b) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$
 (c) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (d) None of these
53. Domain of $\tan\theta =$
 (a) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$ (b) $\mathbb{R}$
 (c) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (d) $n\pi, n \in \mathbb{Z}$

54. Domain $\cot \theta =$
 (a) $\theta \in \mathbb{R}$ (b) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$
 (c) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (d) $\mathbb{R} - \{0\}$
55. Domain of $\sec \theta =$
 (a) $\theta \in \mathbb{R}$ (b) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$
 (c) $\mathbb{R} - (1, 1)$ (d) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$
56. Domain of $\operatorname{cosec} \theta =$
 (a) $\theta \in \mathbb{R}$ (b) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$
 (c) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (d) $\mathbb{R} - [1, 1]$
57. Domain of $\sin^2 \theta + \cos^2 \theta = 1$
 (a) $\theta \in \mathbb{R}$ (b) $\theta \in \mathbb{R}$ but $\theta \neq n\pi, n \in \mathbb{Z}$
 (c) $\theta \in \mathbb{R}$ but $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ (d) $\mathbb{R} - [1, 1]$
58. $\sec \theta \operatorname{cosec} \theta \sin \theta \cos \theta =$
 (a) 1 (b) 0 (c) $\sin \theta$ (d) $\cos \theta$
59. $(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) =$
 (a) 1 (b) $\sec^2 \theta$ (c) $\tan^2 \theta$ (d) $1 - 2 \tan^2 \theta$
60. $\frac{1 - \sin \theta}{\cos \theta} =$
 (a) $\frac{\cos \theta}{1 - \sin \theta}$ (b) $\frac{\cos \theta}{1 + \sin \theta}$ (c) $\frac{\sin \theta}{1 - \cos \theta}$ (d) $\frac{\sin \theta}{1 + \cos \theta}$
61. Which of the following is not quadrantal angle
 (a) 90° (b) -90° (c) -180° (d) 210°
62. Which of the following is quadrantal angle
 (a) 300° (b) -90° (c) -250° (d) 210°
63. Which of the following is quadrantal angle
 (a) -180° (b) -90° (c) -270° (d) All of these

Chapter - 10

Multiple Choice Questions

(Encircle the correct answer choice)

1. Distance between the points $P_1(x_1, y_1)$ & $P_2(x_2, y_2)$ is:
 (a) $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ (b) $d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$
 (c) $d = \sqrt{(x_1 - x_2)^2 + (y_2 - y_1)^2}$ (d) All of these

2. Distance between the points A (3, 8) & (5, 6) is:
 (a) $2\sqrt{2}$ (b) 3 (c) 4 (d) $\sqrt{2}$
3. Fundamental law of Trigonometry is, $\cos(\alpha - \beta) =$
 (a) $\cos \alpha \cos \beta + \sin \alpha \sin \beta$ (b) $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
 (c) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ (d) $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
4. $\cos(\alpha - \beta)$ is equal to
 (a) $\cos \alpha \cos \beta + \sin \alpha \sin \beta$ (b) $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
 (c) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ (d) $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
5. $\cos(\alpha + \beta)$ is equal to
 (a) $\cos \alpha \cos \beta + \sin \alpha \sin \beta$ (b) $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
 (c) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ (d) $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
6. $\sin(\alpha - \beta)$ is equal to
 (a) $\cos \alpha \cos \beta + \sin \alpha \sin \beta$ (b) $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
 (c) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ (d) $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
7. $\sin(\alpha + \beta)$ is equal to
 (a) $\cos \alpha \cos \beta + \sin \alpha \sin \beta$ (b) $\cos \alpha \cos \beta - \sin \alpha \sin \beta$
 (c) $\sin \alpha \cos \beta + \cos \alpha \sin \beta$ (d) $\sin \alpha \cos \beta - \cos \alpha \sin \beta$
8. $\cos\left(\frac{\pi}{2} - \beta\right)$ is equal to
 (a) $\cos \beta$ (b) $-\cos \beta$ (c) $\sin \beta$ (d) $-\sin \beta$
9. $\cos\left(\beta + \frac{\pi}{2}\right)$ is equal to
 (a) $\cos \beta$ (b) $-\cos \beta$ (c) $\sin \beta$ (d) $-\sin \beta$
10. $\sin\left(\beta - \frac{\pi}{2}\right)$ is equal to
 (a) $\cos \beta$ (b) $-\cos \beta$ (c) $\sin \beta$ (d) $-\sin \beta$
11. $\cos(2\pi - \theta)$ is equal to
 (a) $\cos \theta$ (b) $-\cos \theta$ (c) $\sin \theta$ (d) $-\sin \theta$
12. $\sin(2\pi - \theta)$ is equal to
 (a) $\cos \theta$ (b) $-\cos \theta$ (c) $\sin \theta$ (d) $-\sin \theta$
13. $\tan(\alpha + \beta)$ is equal to
 (a) $\frac{\tan \alpha - \tan \beta}{1 - \tan \alpha \tan \beta}$ (b) $\frac{\tan \alpha + \tan \beta}{1 + \tan \alpha \tan \beta}$
 (c) $\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$ (d) $\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$
14. $\tan(\alpha - \beta)$ is equal to
 (a) $\frac{\tan \alpha - \tan \beta}{1 - \tan \alpha \tan \beta}$ (b) $\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$
 (c) $\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$ (d) $\frac{\tan \alpha + \tan \beta}{1 + \tan \alpha \tan \beta}$

15. Angles associated with basic angles of measure θ to a right angle or its multiple are called:
(a) Conterminal angles (b) Angle in standard positions
(c) Allied angles (d) Obtuse angles
16. $\sin\left(\frac{\pi}{2} - \theta\right)$ is equal to
(a) $\cos \theta$ (b) $\sin \theta$ (c) $-\cos \theta$ (d) $-\sin \theta$
17. $\sin\left(\frac{\pi}{2} + \theta\right)$ is equal to
(a) $\cos \theta$ (b) $\sin \theta$ (c) $-\cos \theta$ (d) $-\sin \theta$
18. $\cos\left(\frac{\pi}{2} - \theta\right)$ is equal to
(a) $\cos \theta$ (b) $\sin \theta$ (c) $-\cos \theta$ (d) $-\sin \theta$
19. $\cos\left(\frac{\pi}{2} + \theta\right)$ is equal to
(a) $\cos \theta$ (b) $\sin \theta$ (c) $-\cos \theta$ (d) $-\sin \theta$
20. $\tan\left(\frac{\pi}{2} - \theta\right)$ is equal to
(a) $\cot \theta$ (b) $\tan \theta$ (c) $-\cot \theta$ (d) $-\tan \theta$
21. $\tan\left(\frac{\pi}{2} + \theta\right)$ is equal to
(a) $\cot \theta$ (b) $\tan \theta$ (c) $-\cot \theta$ (d) $-\tan \theta$
22. $\sin(\pi - \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
23. $\sin(\pi + \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
24. $\cos(\pi - \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
25. $\cos(\pi + \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
26. $\tan(\pi - \theta)$ is equal to
(a) $\tan \theta$ (b) $-\cot \theta$ (c) $-\tan \theta$ (d) $\cot \theta$
27. $\tan(\pi + \theta)$ is equal to
(a) $\tan \theta$ (b) $-\cot \theta$ (c) $-\tan \theta$ (d) $\cot \theta$
28. $\sin\left(\frac{3\pi}{2} - \theta\right)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $\sin \theta$ (d) $-\cos \theta$
29. $\sin\left(\frac{3\pi}{2} + \theta\right)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $\sin \theta$ (d) $-\cos \theta$

30. $\cos \left(\frac{3\pi}{2} + \theta \right)$ is equal to
(a) $\sin \alpha$ (b) $\cos \theta$ (c) $\sin \theta$ (d) $-\cos \theta$
31. $\cos \left(\frac{3\pi}{2} - \theta \right)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $\sin \theta$ (d) $-\cos \theta$
32. $\tan \left(\frac{3\pi}{2} + \theta \right)$ is equal to
(a) $\tan \theta$ (b) $\cot \theta$ (c) $-\tan \theta$ (d) $-\cot \theta$
33. $\tan \left(\frac{3\pi}{2} - \theta \right)$ is equal to
(a) $\tan \theta$ (b) $\cot \theta$ (c) $-\tan \theta$ (d) $-\cot \theta$
34. $\sin (2\pi - \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
35. $\sin (2\pi + \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
36. $\cos (2\pi + \theta)$ is equal to
(a) $\sin \theta$ (b) $\cos \theta$ (c) $-\sin \theta$ (d) $-\cos \theta$
37. $\tan (2\pi - \theta)$ is equal to
(a) $\tan \theta$ (b) $\cot \theta$ (c) $-\tan \theta$ (d) $-\cot \theta$
38. $\tan (2\pi + \theta)$ is equal to
(a) $\tan \theta$ (b) $\cot \theta$ (c) $-\tan \theta$ (d) $-\cot \theta$
39. $\cos 315^\circ$ is equal to
(a) 1 (b) 0 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$
40. $\sin 540^\circ$ is equal to
(a) 1 (b) 0 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{\sqrt{3}}{2}$
41. $\tan (-135^\circ)$ is equal to
(a) 1 (b) 0 (c) $\frac{1}{\sqrt{3}}$ (d) -1
42. $\sec (-300^\circ)$ is equal to
(a) 1 (b) 2 (c) 0 (d) -1
43. $\sin (180 + \alpha) \cdot \sin (90 - \alpha)$ is equal to
(a) $\sin \alpha \cos \alpha$ (b) $-\sin \alpha \cos \alpha$ (c) $\cos \gamma$ (d) $-\cos \gamma$
44. If α, β and γ are the angles of ΔABC , then $\sin (\alpha + \beta)$ is equal to
(a) $\sin \gamma$ (b) $-\sin \gamma$ (c) $\cos \gamma$ (d) $-\cos \gamma$

45. If α , β and γ are the angles of ΔABC , then $\cos \frac{(\alpha + \beta)}{2} =$
 (a) $\sin \frac{\pi}{2}$ (b) $-\sin \frac{\pi}{2}$ (c) $\cos \frac{\pi}{2}$ (d) $-\cos \frac{\pi}{2}$
46. If α , β and γ are the angles of ΔABC , then $\cos(\alpha + \beta)$ is equal to
 (a) $\sin \gamma$ (b) $-\sin \gamma$ (c) $\cos \gamma$ (d) $-\cos \gamma$
47. $\frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ} =$
 (a) $\tan 56^\circ$ (b) $\tan 34^\circ$ (c) $\cot 56^\circ$ (d) $\cot 34^\circ$
48. $\sin 2\alpha$ is equal to
 (a) $\cos^2 \alpha - \sin^2 \alpha$ (b) $1 + \cos^2 2\alpha$
 (c) $2 \sin \alpha \cos \alpha$ (d) $2 \sin 2\alpha \cos 2\alpha$
49. $\cos 2\alpha$ is equal to
 (a) $\cos^2 \alpha - \sin^2 \alpha$ (b) $2\cos^2 \alpha - 1$
 (c) $1 - 2 \sin^2 \alpha$ (d) all of these
50. $\tan 2\alpha$ is equal to
 (a) $\frac{2 \tan \alpha}{1 + \tan^2 \alpha}$ (b) $\frac{2 \tan \alpha}{1 - \tan^2 \alpha}$
 (c) $\frac{2 \tan^2 \alpha}{1 - \tan^2 \alpha}$ (d) $\frac{2 \tan^2 \alpha}{1 + \tan^2 \alpha}$
51. $\cos \frac{\alpha}{2} =$
 (a) $\pm \sqrt{\frac{1 + \sin \alpha}{2}}$ (b) $\pm \sqrt{\frac{1 - \cos \alpha}{2}}$ (c) $\pm \sqrt{\frac{1 + \cos \alpha}{2}}$ (d) $\pm \sqrt{\frac{1 - \sin \alpha}{2}}$
52. $\sin \frac{\alpha}{2}$ is equal to
 (a) $\pm \sqrt{\frac{1 + \sin \alpha}{2}}$ (b) $\pm \sqrt{\frac{1 - \cos \alpha}{2}}$ (c) $\pm \sqrt{\frac{1 + \cos \alpha}{2}}$ (d) $\pm \sqrt{\frac{1 - \sin \alpha}{2}}$
53. $\sin 3\alpha$ is equal to
 (a) $3 \sin \alpha - 4 \sin^3 \alpha$ (b) $3 \sin \alpha + 4 \sin^3 \alpha$
 (c) $4 \sin \alpha - 3 \sin^3 \alpha$ (d) $4 \sin \alpha + 3 \sin^3 \alpha$
54. $\cos 3\alpha$ is equal to
 (a) $3 \cos \alpha - 4 \cos^3 \alpha$ (b) $3 \cos^3 \alpha + 4 \cos \alpha$
 (c) $4 \cos^3 \alpha - 3 \cos \alpha$ (d) $4 \cos^3 \alpha + 4 \cos \alpha$
55. $\frac{1 - \cos \alpha}{\sin \alpha} =$
 (a) $\tan \frac{\alpha}{2}$ (b) $\cos \frac{\alpha}{2}$ (c) $\sin \frac{\alpha}{2}$ (d) $\sec \frac{\alpha}{2}$

56. $(\sin \alpha + \sin \beta)$ is equal to

- (a) $2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (b) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$
 (c) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (d) $-2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$

57. $\sin \alpha - \sin \beta =$

- (a) $2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (b) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$
 (c) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (d) $-2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$

58. $\cos \alpha + \cos \beta =$

- (a) $2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (b) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$
 (c) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (d) $-2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$

59. $\cos \alpha - \cos \beta =$

- (a) $2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (b) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$
 (c) $2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)$ (d) $-2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)$

60. $2 \sin 7\theta \cos 3\theta =$

- (a) $\sin 10\theta + \sin 4\theta$ (b) $\sin 5\theta + \sin 2\theta$
 (c) $\cos 10\theta + \cos 4\theta$ (d) $\cos 5\theta - \cos 2\theta$

61. $2 \cos 5\theta \sin 3\theta$ is equal to

- (a) $\sin 8\theta - \sin 2\theta$ (b) $\sin 8\theta + \sin 2\theta$
 (c) $\sin 4\theta - \sin \theta$ (d) $\sin 4\theta + \sin \theta$

62. $2 \sin 7\theta \sin 2\theta$ is equal to

- (a) $\cos 5\theta - \cos 9\theta$ (b) $\cos 9\theta - \cos 5\theta$
 (c) $\sin 9\theta + \sin 5\theta$ (d) $\sin 9\theta + \sin 5\theta$

63. $\sin 12^\circ \sin 46^\circ$ is equal to

- (a) $\frac{1}{2} (\cos 34^\circ - \cos 58^\circ)$ (b) $\frac{1}{2} (\cos 58^\circ - \cos 34^\circ)$
 (c) $(\cos 58^\circ - \cos 34^\circ)$ (d) $\frac{1}{2} (\cos 58^\circ + \cos 34^\circ)$

64. Which is the allied angle:

- (a) $90^\circ + \theta$ (b) $60^\circ + \theta$ (c) $45^\circ + \theta$ (d) $30^\circ + \theta$

65. The value of $\sin 420^\circ$ is:

- (a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $-\frac{1}{2}$ (d) $-\frac{\sqrt{3}}{2}$

66. $2 \sin x \cos x$ is equal to:
 (a) $\sin x$ (b) $\sin 2x$ (c) $\sin x/2 \cos x/2$ (d) none of these
67. The value of $\cos(\alpha - 2\pi)$ is equal to:
 (a) $-\cos \alpha$ (b) $-\sin \alpha$ (c) $\cos \alpha$ (d) $\sin \alpha$
68. The value of $\sin 7\pi$ is equal to:
 (a) 0 (b) 1 (c) -1 (d) $1/2$

Chapter - 11

Multiple Choice Questions

Encircle the correct answer choice)

- Domain of $y = \sin x$ is
 (a) $-x < x < x$ (b) $-1 \leq x \leq 1$
 (c) $-\infty < x < \infty, x \neq n\pi, n \in \mathbb{Z}$ (d) $x \geq 1, x \leq -1$
- Domain of $y = \cos x$ is
 (a) $-\infty < x < \infty$ (b) $-1 \leq x \leq 1$
 (c) $-\infty < x < \infty, x \neq n\pi, n \in \mathbb{Z}$ (d) $x \geq 1, x \leq -1$
- Domain of $y = \tan x$ is
 (a) $-\infty < x < \infty$ (b) $-\infty < x < \infty, x \neq n\pi, n \in \mathbb{Z}$
 (c) $-\infty < x < \infty, x \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$ (d) $-\pi \leq x \leq \pi$
- Domain of $y = \cot x$ is
 (a) $-\infty < x < \infty$ (b) $-\infty < x < \infty, x \neq n\pi, n \in \mathbb{Z}$
 (c) $-\infty < x < \infty, x \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$ (d) $-\pi \leq x \leq \pi$
- Domain of $y = \sec x$ is
 (a) $-\infty < x < \infty$ (b) $-\infty < x < \infty, x \neq n\pi, n \in \mathbb{Z}$
 (c) $-\infty < x < \infty, x \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$ (d) $-\pi \leq x \leq \pi$
- Domain of $y = \operatorname{cosec} x$ is
 (a) $-\infty < x < \infty$ (b) $-\infty < x < \infty, x \neq n\pi, n \in \mathbb{Z}$
 (c) $-\infty < x < \infty, x \neq \frac{(2n+1)\pi}{2}, n \in \mathbb{Z}$ (d) $-\pi \leq x \leq \pi$
- Range of $y = \sin x$ is
 (a) $\mathbb{R}$ (b) $-1 \leq y \leq 1$ (c) $(-\infty, 1) \cup (1, \infty)$ (d) $-1 < y < 1$
- Range of $y = \cos x$ is
 (a) $\mathbb{R}$ (b) $[-1, 1]$ (c) $-1 < y < 1$ (d) $(-\infty, 1) \cup (1, \infty)$
- Range of $y = \tan x$ is
 (a) $(-\infty, \infty)$ (b) $[-1, 1]$ (c) $\mathbb{Q}$ (d) $\mathbb{R} - \{0\}$
- Range of $y = \cot x$ is
 (a) $\mathbb{R}$ (b) $\mathbb{R} - [-1, 1]$ (c) $\mathbb{R} - \{0\}$ (d) $\mathbb{Z}$

11. Range of $y = \sec x$ is
(a) R (b) $y \geq 1$ or $y \leq -1$
(c) $-1 \leq y \leq 1$ (d) $R - [-1, 1]$
12. Range of $y = \operatorname{Cosec} x$ is
(a) R (b) $y \geq 1$ or $y \leq -1$
(c) $-1 \leq y \leq 1$ (d) $R - [-1, 1]$
13. Smallest +ve number which when added to the original circular measure of the angle gives the same value of the function is called
(a) Domain (b) Range (c) co domain (d) period
14. Period of $\sin \theta$ is
(a) π (b) 2π (c) -2π (d) $\frac{\pi}{2}$
15. Period of $\operatorname{cosec} \theta$ is
(a) π (b) 2π (c) -2π (d) $\frac{3\pi}{2}$
16. Period of $\tan \theta$ is
(a) π (b) 2π (c) -2π (d) $\frac{3\pi}{2}$
17. Period of $\cot \theta$ is
(a) π (b) 2π (c) -2π (d) $\frac{3\pi}{2}$
18. Period of $\sec \theta$ is
(a) π (b) 2π (c) -2π (d) $\frac{3\pi}{2}$
19. Period of $\cos \theta$ is
(a) π (b) 2π (c) -2π (d) $\frac{3\pi}{2}$
20. period of $\sin 3x$ is
(a) π (b) 2π (c) $\frac{2\pi}{3}$ (d) 6π
21. Period of $\cos 2x$ is
(a) 2π (b) π (c) 4π (d) $\frac{\pi}{2}$
22. Period of $\tan 4x$ is
(a) $\frac{\pi}{4}$ (b) 4π (c) 8π (d) $\frac{\pi}{2}$
23. Period of $\cot 3x$ is
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) 3π

24. Period of $\sec 2x$ is
(a) π (b) 2π (c) $\frac{\pi}{2}$ (d) 4π
25. Period of $\operatorname{cosec} 3x$ is
(a) π (b) $\frac{\pi}{3}$ (c) $\frac{2\pi}{3}$ (d) 3π
26. Period of $\sin \frac{x}{3}$ is
(a) 2π (b) $\frac{2\pi}{3}$ (c) 6π (d) 3π
27. Period of $\cos \frac{x}{6}$ is
(a) 12π (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) 3π
28. Period of $\cot \frac{x}{2}$
(a) 2π (b) $\frac{\pi}{2}$ (c) π (d) $\pi/4$
29. Period of $3 \cos \frac{x}{5}$ is
(a) $\frac{10\pi}{3}$ (b) $\frac{6\pi}{5}$ (c) 10π (d) $\frac{5\pi}{3}$
30. Period of $2 \operatorname{cosec} \frac{x}{4}$ is
(a) 2π (b) 4π (c) $\frac{\pi}{2}$ (d) 8π
31. Period of $3 \tan \frac{x}{7}$ is
(a) $\frac{7\pi}{3}$ (b) 7π (c) $\frac{14\pi}{3}$ (d) 14π
32. The graph of trigonometric functions have
(a) Breaks segments (b) sharp corners
(c) straight line segments (d) smooth curves
33. The graph of function $y = \sin 2x$, will be between the lines
(a) $x = 1$ & $x = -1$ (b) $y = 1$ & $y = -1$
(c) $x = 2$ & $x = -2$ (d) $y = 2$ & $y = -2$
34. The graph of sine function in the interval $[0, 2\pi]$ is called
(a) its period (b) its domain (c) its continuity (d) one cycle
35. The graph of function $y = 2\sin x$, will be between the lines
(a) $x = 1$ & $x = -1$ (b) $y = 1$ & $y = -1$
(c) $x = 2$ & $x = -2$ (d) $y = 2$ & $y = -2$

36. The trigonometric functions repeat their values after adding or subtracting 2π in basic angle x . This behavior is called
 (a) period (b) periodicity (c) continuity (d) range

Chapter - 12

Multiple Choice Questions

(Encircle the correct answer choice)

- A "Triangle" has:
 (a) Two elements (b) 3 Elements
 (c) 4 Elements (d) 6 Elements
- At the top of a cliff 80m high the angle of depression of a boat is α . If the distance between the boat and foot of cliff is $80\sqrt{3}$ m, then angle α is
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{3\pi}{4}$
- When we look an object above the horizontal ray, the angle formed is
 (a) Angle of Elevation (b) Angle of depression
 (c) Angle of incidence (d) Angle of reflection
- When we look an object below the horizontal ray, the angle formed is
 (a) Angle of Elevation (b) Angle of depression
 (c) Angle of incidence (d) Angle of reflection
- A Triangle which is not right is called:
 (a) Oblique triangle (b) Isosceles triangle
 (c) Scalene triangle (d) Right Isosceles triangle
- To solve an oblique triangle, we use:
 (a) Law of sines (b) Law of cosines
 (c) Law of tangents (d) All of these
- In any triangle ABC, $\frac{b^2 + c^2 - a^2}{2bc} =$
 (a) $\cos \alpha$ (b) $\sin \alpha$ (c) $\cos \beta$ (d) $\cos \gamma$
- Which can be reduced to Pythagoras theorem:
 (a) Law of sines (b) Law of cosines
 (c) Law of tangents (d) Half angle formulas
- In any triangle ABC, if $\beta = 90^\circ$, then $b^2 = c^2 + a^2 - 2ac \cos \beta$ becomes:
 (a) Law of sin (b) Law of Tangents
 (c) Pythagoras Theorem (d) None of these
- In any triangle ABC, Law of tangent is:
 (a) $\frac{a-b}{a+b} = \frac{\tan(\frac{\alpha-\beta}{2})}{\tan(\frac{\alpha+\beta}{2})}$ (b) $\frac{a-b}{a+b} = \frac{\tan(\frac{\alpha+\beta}{2})}{\tan(\frac{\alpha-\beta}{2})}$
 (c) $\frac{a-b}{a+b} = \frac{\tan(\frac{\alpha-\beta}{2})}{\tan(\frac{\alpha+\beta}{2})}$ (d) $\frac{a-b}{a+b} = \frac{\tan(\frac{\alpha+\beta}{2})}{\tan(\frac{\alpha-\beta}{2})}$

11. In any triangle ABC, $\sqrt{\frac{(s-b)(s-c)}{bc}} =$

- (a) $\sin \frac{\alpha}{2}$ (b) $\cos \frac{\alpha}{2}$ (c) $\sin \frac{\beta}{2}$ (d) $\sin \frac{\gamma}{2}$

12. In any triangle ABC, $\sqrt{\frac{(s-a)(s-c)}{ac}}$ is equal to

- (a) $\sin \frac{\alpha}{2}$ (b) $\cos \frac{\alpha}{2}$ (c) $\sin \frac{\beta}{2}$ (d) $\sin \frac{\gamma}{2}$

13. In any triangle ABC, $\sqrt{\frac{(s-a)(s-b)}{ab}} =$

- (a) $\sin \frac{\alpha}{2}$ (b) $\cos \frac{\alpha}{2}$ (c) $\sin \frac{\beta}{2}$ (d) $\sin \frac{\gamma}{2}$

14. In any triangle ABC, $\cos \frac{\alpha}{2}$ is equal to

- (a) $\sqrt{\frac{s(s-a)}{ab}}$ (b) $\sqrt{\frac{s(s-b)}{ac}}$ (c) $\sqrt{\frac{s(s-a)}{bc}}$ (d) $\sqrt{\frac{s(s-c)}{ab}}$

15. In any triangle ABC, $\cos \frac{\beta}{2}$ is equal to

- (a) $\sqrt{\frac{s(s-a)}{ab}}$ (b) $\sqrt{\frac{s(s-b)}{ac}}$ (c) $\sqrt{\frac{s(s-a)}{bc}}$ (d) $\sqrt{\frac{s(s-c)}{ab}}$

16. In any triangle ABC, $\cos \frac{\gamma}{2}$ is equal to

- (a) $\sqrt{\frac{s(s-a)}{ab}}$ (b) $\sqrt{\frac{s(s-b)}{ac}}$ (c) $\sqrt{\frac{s(s-a)}{bc}}$ (d) $\sqrt{\frac{s(s-c)}{ab}}$

17. In any triangle ABC, with usual notations, s is equal to

- (a) $a+b+c$ (b) $\frac{a+b+c}{2}$ (c) $\frac{a+b+c}{3}$ (d) $\frac{abc}{2}$

18. $\sqrt{\frac{s(s-a)}{(s-b)(s-c)}} =$

- (a) $\sin \frac{\alpha}{2}$ (b) $\cos \frac{\alpha}{2}$ (c) $\tan \frac{\alpha}{2}$ (d) $\cot \frac{\alpha}{2}$

19. $\sqrt{\frac{s(s-b)}{(s-a)(s-c)}} =$

- (a) $\sin \frac{\beta}{2}$ (b) $\cos \frac{\beta}{2}$ (c) $\tan \frac{\beta}{2}$ (d) $\cot \frac{\beta}{2}$

20. In any triangle ABC, $\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$ is equal to
 (a) $\sin \frac{\gamma}{2}$ (b) $\cos \frac{\gamma}{2}$ (c) $\tan \frac{\gamma}{2}$ (d) $\cot \frac{\gamma}{2}$
21. In any triangle ABC, $\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$ is equal to
 (a) $\sin \frac{\gamma}{2}$ (b) $\cos \frac{\gamma}{2}$ (c) $\tan \frac{\gamma}{2}$ (d) $\cot \frac{\gamma}{2}$
22. In any triangle ABC, $\sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$
 (a) $\tan \frac{\beta}{2}$ (b) $\tan \frac{\gamma}{2}$ (c) $\tan \frac{\gamma}{2}$ (d) $\sec \frac{\gamma}{2}$
23. In any triangle ABC, $\sqrt{\frac{(s+a)(s+b)}{s(s+c)}}$ =
 (a) $\sin \frac{\gamma}{2}$ (b) $\cos \frac{\gamma}{2}$ (c) $\tan \frac{\gamma}{2}$ (d) None of these
24. We can solve an oblique triangle, if:
 (a) One side and two angles are known (b) Three sides are known
 (c) Two sides and their included angles are known (d) All (a), (b) and (c)
25. To solve an oblique Triangle when measure of three sides are given, we can use:
 (a) Hero Formula (b) Law of Cosines
 (c) Law of Tangents (d) Pythagoras theorem
26. The smallest angle of ΔABC , when $a = 37.34$, $b = 3.24$, and $c = 35.06$ is
 (a) α (b) β (c) γ (d) cannot be determined
27. Area of Triangle in terms of measure of two sides and their included angle is:
 (a) $\frac{1}{2}bc \sin \alpha$ (b) $\frac{1}{2}ca \sin \beta$ (c) $\frac{1}{2}ab \sin \gamma$ (d) All of these
28. In any triangle ABC, Area of Triangle is:
 (a) $bc \sin \alpha$ (b) $\frac{1}{2}ca \sin \alpha$ (c) $\frac{1}{2}ab \sin \gamma$ (d) $\frac{1}{2}ab \sin \beta$
29. Area of Triangle in terms of measure of one side and two angles is:
 (a) $\frac{1}{2} \frac{a^2 \sin \beta \sin \gamma}{\sin \alpha}$ (b) $\frac{1}{2} \frac{b^2 \sin \alpha \sin \gamma}{\sin \beta}$
 (c) $\frac{1}{2} \frac{c^2 \sin \alpha \sin \beta}{\sin \gamma}$ (d) All of these

30. In any triangle ABC, Hero's formula is

(a) $\Delta = s(s-a)(s-b)(s-c)$ (b) $\Delta = \sqrt{(s-a)(s-b)(s-c)}$

(c) $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ (d) $\Delta = \frac{a+b+c}{2}$

31. In any triangle ABC, with usual notations, which one of them is not true

(a) $\Delta = \frac{1}{2}ab \sin \gamma$ (b) $\Delta = \frac{1}{2}bc \sin \alpha$

(c) $\Delta = \frac{1}{2} \frac{a \sin \beta \sin \gamma}{\sin \alpha}$ (d) $\Delta^2 = s(s-a)(s-b)(s-c)$

32. The circle passing through the three vertices of a Triangle is called

(a) Circum Circle (b) In-circle (c) Ex-centre (d) Escribed circle

33. The point of intersection of the right bisectors of the sides of the Triangle is called:

(a) circum centre (b) In centre
(c) Escribed centre (d) ortho centre

34. Radius of the circle which passes through the vertices of a Triangle is:

(a) Circum Radius (b) In-Radius
(c) e-Radius (d) Diameter

35. In any triangle ABC, with usual notations, $\frac{a}{2 \sin \alpha} =$

(a) r (b) r_1 (c) R (d) Δ

36. In any triangle ABC, with usual notations, $\frac{b}{\sin \beta} =$

(a) $2r$ (b) $2r_1$ (c) $2R$ (d) ΔS

37. In any triangle ABC, with usual notations, $\sin \gamma =$

(a) R (b) $\frac{c}{2R}$ (c) $\frac{2R}{c}$ (d) $\frac{R}{2}$

38. In any triangle ABC, with usual notations, $R =$

(a) $\frac{abc}{\Delta}$ (b) $\frac{abc}{4\Delta}$ (c) $\frac{4\Delta}{abc}$ (d) $\frac{\Delta}{abc}$

39. In any triangle ABC, with usual notations, $abc =$

(a) R (b) R_s (c) $4R\Delta$ (d) $\frac{\Delta}{s}$

40. The circle drawn inside a Triangle touching its three sides internally is

(a) Inscribed circle (b) unit circle
(c) circum circle (d) Escribed circle

41. The point of intersection of the bisectors of angles of the Triangle is:

(a) In centre (b) e-centre
(c) circum centre (d) Ex-centre

42. In Radius is denoted by
 (a) R (b) r (c) r_1 (d) s
43. In any triangle ABC, with usual notations, in radius r is equal to
 (a) $\frac{s}{\Delta}$ (b) $\frac{\Delta}{s}$ (c) $s\Delta$ (d) $\frac{a}{2\sin\alpha}$
44. A circle which touches one side of the Triangle externally and other two produced sides internally is
 (a) Escribed circle (b) Ex-circle (c) e-circle (d) all of these
45. The point where the internal bisector of one and the external bisector of the other two angles of the Triangle meet is called:
 (a) Escribed centre (b) Ex- centre (c) e-centre (d) all of these
46. In any triangle ABC, with usual notations, $\frac{\Delta}{s-a}$:
 (a) R (b) r (c) r_1 (d) $\sin\alpha$
47. In any triangle ABC, with usual notations, r_3 is equal to
 (a) $\frac{\Delta}{s-a}$ (b) $\frac{\Delta}{s-b}$ (c) $\frac{\Delta}{s-c}$ (d) $\frac{s-a}{\Delta}$
48. In any triangle ABC, with usual notations, $r : R : r_1$
 (a) 1: 2: 3 (b) 3: 2: 1 (c) 1:3: 2 (d) 1:1: 1
49. In any triangle ABC, with usual notations, Law of Sine is:
 (a) $\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$ (b) $\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$
 (c) $a : \sin\alpha = b : \sin\beta = c : \sin\gamma$ (d) All of these
50. The area of triangle ABC is
 (a) $\frac{1}{2}bc\sin\beta$ (b) $\frac{1}{2}bcsin\gamma$ (c) $\frac{1}{2}bcsin\alpha$ (d) $\frac{1}{2}bcsin(\alpha + \beta + \gamma)$
51. For a circum circle, R =
 (a) $\frac{abc}{4\Delta}$ (b) $\frac{a}{4s\Delta}$ (c) $\frac{abc}{\Delta}$ (d) $\frac{4\Delta}{abc}$
52. In a triangle ABC if $\beta = 60^\circ$, $\gamma = 15^\circ$, then α equals:
 (a) 90° (b) 180° (c) 150° (d) 105°
53. With usual notation r_3 equals
 (a) $\frac{\Delta}{s-a}$ (b) $\frac{\Delta}{s-b}$ (c) $\frac{\Delta}{s-c}$ (d) $\frac{\Delta}{s+a}$
54. With usual notations, $\frac{\Delta}{s-a}$ is equal to:
 (a) r (b) r_1 (c) r_2 (d) r_3
55. With usual notation r_2 is equal to:
 (a) $\frac{\Delta}{s-b}$ (b) $\frac{\Delta}{s}$ (c) $\frac{s-b}{\Delta}$ (d) $\frac{s}{\Delta}$

Chapter - 13**Multiple Choice Questions***(Encircle the correct answer choice)*

Note: Here we are dealing with principal function or capital function i.e.

1. Instead of $\sin x$, we use $\text{Sin } x$.
Instead of $\cos x$, we use $\text{Cos } x$.
Instead of $\tan x$ we use $\text{Tan } x$ etc. While in chapter 11, General functions were discussed and symbols $\sin x$, $\tan x$ etc. were used.
2. Here we are restricting the domain to make the function (1-1), so that its inverse is to be calculated.
3. Inverse of general Trigonometric functions does not exist. It exists only when function is (1-1), to make the function (1-1), we restrict the domain of the function and we call the function principal or capital functions. We denote the principal functions as:

$$\begin{array}{lll} y = \text{Sin } x, & y = \text{Cos } x, & y = \text{Sec } x \\ y = \text{Cosec } x, & y = \text{Tan } x, & y = \text{Cot } x \end{array}$$

1. If $y = \text{Sin } x$, then Domain is

$$\begin{array}{ll} \text{(a)} -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} & \text{(b)} 0 \leq x \leq \pi \\ \text{(c)} [0, \pi], x \neq \frac{\pi}{2} & \text{(d)} [-\frac{\pi}{2}, \frac{\pi}{2}], x \neq 0 \end{array}$$

2. If $y = \text{Cos } x$, then Domain is

$$\begin{array}{ll} \text{(a)} -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} & \text{(b)} 0 \leq x \leq \pi \\ \text{(c)} [0, \pi], x \neq \frac{\pi}{2} & \text{(d)} [-\frac{\pi}{2}, \frac{\pi}{2}], x \neq 0 \end{array}$$

3. If $y = \text{Sec } x$, then Domain is

$$\begin{array}{ll} \text{(a)} -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} & \text{(b)} 0 \leq x \leq \pi \\ \text{(c)} [0, \pi], x \neq \frac{\pi}{2} & \text{(d)} [-\frac{\pi}{2}, \frac{\pi}{2}], x \neq 0 \end{array}$$

4. If $y = \text{Cosec } x$, then Domain is

$$\begin{array}{ll} \text{(a)} -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} & \text{(b)} 0 \leq x \leq \pi \\ \text{(c)} [0, \pi], x \neq \frac{\pi}{2} & \text{(d)} [-\frac{\pi}{2}, \frac{\pi}{2}], x \neq 0 \end{array}$$

5. If $y = \tan x$, then domain is
(a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (b) $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (c) $0 < x < \pi$ (d) $0 \leq x \leq \pi$
6. If $y = \cot x$, then domain is
(a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (b) $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (c) $0 < x < \pi$ (d) $0 \leq x \leq \pi$
7. If $y = \sin x$, then range is
(a) $-1 \leq y \leq 1$ (b) $(-\infty, +\infty)$ or R (c) $y \leq -1$ or $y \geq 1$ (d) $y < -1$ or $y > 1$
8. If $y = \cos x$, then range is
(a) $-1 \leq y \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $y \leq -1$ or $y \geq 1$ (d) $y < -1$ or $y > 1$
9. If $y = \tan x$, Then range is
(a) $-1 \leq y \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $y \leq -1$ or $y \geq 1$ (d) $y < -1$ or $y > 1$
10. If, $y = \cot x$, then range is
(a) $-1 \leq y \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $y \leq -1$ or $y \geq 1$ (d) $y < -1$ or $y > 1$
11. If $y = \operatorname{cosec} x$, then range is
(a) $-1 \leq y \leq 1$ (b) $(-\infty, +\infty)$ or R (c) $y \leq -1$ or $y \geq 1$ (d) $y < -1$ or $y > 1$
12. If $y = \sec x$, then range is
(a) $-1 \leq y \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $y \leq -1$ or $y \geq 1$ (d) $y < -1$ or $y > 1$
13. If $y = \sin^{-1}x$, then domain is
(a) $-1 \leq x \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $x \geq -1$ or $x \leq 1$ (d) $x \leq -1$ or $x \geq 1$
14. If $y = \cos^{-1}x$, then Domain is
(a) $-1 \leq x \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $x \geq -1$ or $x \leq 1$ (d) $x \leq -1$ or $x \geq 1$
15. If $y = \tan^{-1}x$, then domain is
(a) $-1 \leq x \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $x \geq -1$ or $x \leq 1$ (d) $x \leq -1$ or $x \geq 1$
16. If $y = \cot^{-1}x$, then Domain is
(a) $-1 \leq x \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $x \geq -1$ or $x \leq 1$ (d) $x \leq -1$ or $x \geq 1$
17. If $y = \operatorname{see}^{-1}x$, then Domain is
(a) $-1 \leq x \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $x \geq -1$ or $x \leq 1$ (d) $x \leq -1$ or $x \geq 1$
18. If $y = \operatorname{cosec}^{-1}x$, then Domain is
(a) $-1 \leq x \leq 1$ (b) $(-\infty, +\infty)$ or R
(c) $x \geq -1$ or $x \leq 1$ (d) $x \leq -1$ or $x \geq 1$

19. If $y = \sin^{-1} x$, then range is

- (a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (b) $0 \leq x \leq \pi$ (c) $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (d) $0 < x < \pi$

20. If $y = \cos^{-1} x$, then range is

- (a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (b) $0 \leq x \leq \pi$ (c) $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (d) $0 < x < \pi$

21. If $y = \tan^{-1} x$, then range is

- (a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (b) $0 \leq x \leq \pi$ (c) $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (d) $0 < x < \pi$

22. If $y = \cot^{-1} x$, then range is

- (a) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (b) $0 \leq x \leq \pi$ (c) $-\frac{\pi}{2} < x < \frac{\pi}{2}$ (d) $0 < x < \pi$

23. If $y = \sec^{-1} x$, then range is

- (a) $0 \leq y \leq \pi, y \neq \frac{\pi}{2}$ (b) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$

- (c) $0 < y < \pi$ (d) $-\frac{\pi}{2} < y < \frac{\pi}{2}$

24. If $y = \operatorname{cosec}^{-1} x$, then range is

- (a) $0 \leq y \leq \pi, y \neq \frac{\pi}{2}$ (b) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}, y \neq 0$

- (c) $0 < y < \pi$ (d) $-\frac{\pi}{2} < y < \frac{\pi}{2}$

25. Inverse of a function exist only if it is

- (a) trigonometric function (b) (1-1) function
(c) onto function (d) an into function

26. If $y = \sin^{-1} x$, then which is not true

- (a) $x = \sin y$ (b) domain of Inverse function is value of x
(c) $y = (\sin)^{-1}$ (d) range of Inverse function is value of y

27. $\sin^{-1} x =$

- (a) $\frac{\pi}{2} - \cos^{-1} x$ (b) $\frac{\pi}{2} - \sin^{-1} x$ (c) $\frac{\pi}{2} + \cos^{-1} x$ (d) $\frac{\pi}{2} - \operatorname{cosec}^{-1} x$

28. $\cos^{-1} x = \dots\dots\dots$

- (a) $\frac{\pi}{2} - \cos^{-1} x$ (b) $\frac{\pi}{2} - \sin^{-1} x$ (c) $\frac{\pi}{2} - \sec^{-1} x$ (d) $\frac{\pi}{2} + \cos^{-1} x$

29. $\operatorname{cosec}^{-1} x = \dots\dots\dots$

- (a) $\frac{\pi}{2} - \sec^{-1} x$ (b) $\frac{\pi}{2} - \operatorname{cosec}^{-1} x$ (c) $\frac{\pi}{2} + \operatorname{cosec}^{-1} x$ (d) $\frac{\pi}{2} - \sin^{-1} x$

30. $\sec^{-1}x =$

- (a)
- $\frac{\pi}{2} - \operatorname{cosec}^{-1}x$
- (b)
- $\frac{\pi}{2} - \sec^{-1}x$
- (c)
- $\frac{\pi}{2} - \cos^{-1}x$
- (d)
- $\frac{\pi}{2} + \sec^{-1}x$

31. $\tan^{-1}x =$

- (a)
- $\frac{\pi}{2} - \tan^{-1}x$
- (b)
- $\frac{\pi}{2} - \cot^{-1}x$
- (c)
- $\frac{\pi}{2} + \tan^{-1}x$
- (d)
- $\frac{\pi}{2} + \cot^{-1}x$

32. $\cot^{-1}x = \dots\dots\dots$

- (a)
- $\frac{\pi}{2} - \tan^{-1}x$
- (b)
- $\frac{\pi}{2} - \cot^{-1}x$
- (c)
- $\frac{\pi}{2} + \tan^{-1}x$
- (d)
- $\frac{\pi}{2} + \cot^{-1}x$

33. $\sin(\cos^{-1}\frac{\sqrt{3}}{2}) = \dots\dots\dots$

- (a)
- $\frac{\pi}{6}$
- (b)
- $\frac{1}{2}$
- (c)
- $-\frac{1}{2}$
- (d)
- $\frac{\sqrt{3}}{2}$

34. $\cos(\tan^{-1}0) = \dots\dots\dots$

- (a) 0 (b) 1 (c)
- $\frac{\pi}{2}$
- (d) -1

35. $\sec[\sin^{-1}(-\frac{1}{2})] = \dots\dots\dots$

- (a)
- $\frac{2}{\sqrt{3}}$
- (b)
- $\frac{\sqrt{3}}{2}$
- (c)
- $\frac{1}{2}$
- (d)
- $-\frac{2}{\sqrt{3}}$

36. $\sin^{-1}(\frac{1}{2}) =$

- (a)
- $\frac{\pi}{6}$
- (b)
- $-\frac{\pi}{6}$
- (c)
- $-\frac{\pi}{3}$
- (d)
- $\frac{\pi}{3}$

37. $\cos^{-1}(\frac{1}{2}) = \dots\dots\dots$

- (a)
- $\frac{\pi}{6}$
- (b)
- $-\frac{\pi}{6}$
- (c)
- $-\frac{\pi}{3}$
- (d)
- $\frac{\pi}{3}$

38. $\tan^{-1}(-\frac{1}{3}) = \dots\dots\dots$

- (a)
- $\frac{\pi}{6}$
- (b)
- $-\frac{\pi}{6}$
- (c)
- $-\frac{\pi}{3}$
- (d)
- $\frac{\pi}{3}$

39. $\tan^{-1}(\sqrt{3}) = \dots\dots\dots$

- (a)
- $\frac{\pi}{6}$
- (b)
- $-\frac{\pi}{6}$
- (c)
- $-\frac{\pi}{3}$
- (d)
- $\frac{\pi}{3}$

40. $\cot^{-1}(-1) =$

- (a) $\frac{\pi}{4}$ (b) $-\frac{\pi}{4}$ (c) $\frac{3\pi}{4}$ (d) $-\frac{3\pi}{4}$

41. $\tan \tan^{-1}(-1) =$

- (a) -1 (b) 1 (c) $\frac{\pi}{4}$ (d) $-\frac{\pi}{4}$

42. $\cos(\sin^{-1} \frac{1}{\sqrt{2}}) =$

- (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$

43. $\sec(\cos^{-1} \frac{1}{2}) =$

- (a) 2 (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{2}{\sqrt{3}}$

44. $\tan \cos^{-1} \frac{\sqrt{3}}{2} =$

- (a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

45. $\operatorname{cosec}(\tan^{-1}(-1)) = \dots\dots$

- (a) $\frac{1}{\sqrt{2}}$ (b) $-\frac{1}{\sqrt{2}}$ (c) $\sqrt{2}$ (d) $-\sqrt{2}$

46. $\sin(\sin^{-1} \frac{1}{2}) = \dots\dots$

- (a) $\frac{1}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $\frac{\sqrt{3}}{2}$

47. $\tan(\sin^{-1}(-\frac{1}{2})) = \dots\dots$

- (a) $\sqrt{3}$ (b) $-\sqrt{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $-\frac{1}{\sqrt{3}}$

48. $\sin^{-1}A + \sin^{-1}B$ is equal to

- (a) $\sin^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$ (b) $\sin^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$
(c) $\sin^{-1}(B\sqrt{1-A^2}) - (A\sqrt{1-B^2})$ (d) $\sin^{-1}(AB\sqrt{(1-A^2)(1-B^2)})$

49. $\sin^{-1}A - \sin^{-1}B = \dots\dots$

- (a) $\sin^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$ (b) $\sin^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$
(c) $\sin^{-1}(B\sqrt{1-A^2}) - (A\sqrt{1-B^2})$ (d) $\sin^{-1}(AB\sqrt{(1-A^2)(1-B^2)})$

50. $\cos^{-1} A + \cos^{-1} B = \dots\dots$

(a) $\cos^{-1}(AB - \sqrt{(1-A^2)(1-B^2)})$

(b) $\cos^{-1}(AB + \sqrt{(1-A^2)(1-B^2)})$

(c) $\cos^{-1}(AB + \sqrt{(1+A^2)(1+B^2)})$

(d) $\cos^{-1}(AB - \sqrt{(1+A^2)(1+B^2)})$

51. $\cos^{-1} A - \cos^{-1} B = \dots\dots\dots$

(a) $\cos^{-1}(AB - \sqrt{(1-A^2)(1-B^2)})$

(b) $\cos^{-1}(AB + \sqrt{(1-A^2)(1-B^2)})$

(c) $\cos^{-1}(AB + \sqrt{(1+A^2)(1+B^2)})$

(d) $\cos^{-1}(AB - \sqrt{(1+A^2)(1+B^2)})$

52. $\tan^{-1} A + \tan^{-1} B =$

(a) $\tan^{-1}\left(\frac{A+B}{1+AB}\right)$

(b) $\tan^{-1}\left(\frac{A+B}{1-AB}\right)$

(c) $\tan^{-1}\left(\frac{A-B}{1-AB}\right)$

(d) $\tan^{-1}\left(\frac{A+B}{1+AB}\right)$

53. $\tan^{-1} A - \tan^{-1} B =$

(a) $\tan^{-1}\left(\frac{A-B}{1+AB}\right)$

(b) $\tan^{-1}\left(\frac{A+B}{1-AB}\right)$

(c) $\tan^{-1}\left(\frac{A-B}{1-AB}\right)$

(d) $\tan^{-1}\left(\frac{A+B}{1+AB}\right)$

54. $2 \tan^{-1} A =$

(a) $\tan^{-1}\left(\frac{A}{1-A^2}\right)$

(b) $\tan^{-1}\left(\frac{2A}{1-A^2}\right)$

(c) $\tan^{-1}\left(\frac{2A}{1+A^2}\right)$

(d) $\tan^{-1}\left(\frac{A}{1+A^2}\right)$

55. $\sin^{-1}(-x) = \dots\dots\dots$

(a) $-\sin^{-1} x$

(b) $\sin^{-1} x$

(c) $\pi - \sin^{-1} x$

(d) $\pi - \sin x$

56. $\cos^{-1}(-x)$

(a) $\cos^{-1} x$

(b) $\sin^{-1} x$

(c) $\pi - \cos^{-1} x$

(d) $-\cos^{-1} x$

57. $\tan^{-1}(-x) = \dots\dots\dots$

(a) $-\tan^{-1} x$

(b) $\pi - \tan^{-1} x$

(c) $\cot^{-1} x$

(d) $\tan^{-1} x$

58. $2 \sin^{-1} A = \dots\dots\dots$

(a) $\sin^{-1}(2A \sqrt{1-A^2})$

(b) $\sin^{-1}(A \sqrt{1-A^2})$

(c) $\sin^{-1}(2A \sqrt{1+A^2})$

(d) $\cos^{-1}(2A \sqrt{1-A^2})$

59. $2 \cos^{-1} A =$

(a) $\cos^{-1}(2A^2-1)$

(b) $\cos^{-1}(1-2A^2)$

(c) $\cos^{-1}(2A-1)$

(d) $\cos^{-1}(2A^2+1)$

60. $\operatorname{cosec}^{-1}(-x) = \dots\dots\dots$

(a) $-\operatorname{cosec}^{-1} x$

(b) $\sin^{-1} x$

(c) $\pi - \operatorname{cosec}^{-1} x$

(d) $\pi - \sin^{-1} x$

61. $\sec^{-1}(-x)$
(a) $\cos^{-1}x$ (b) $\sec^{-1}x$ (c) $\pi - \sec^{-1}x$ (d) $-\sec^{-1}x$
62. $\cot^{-1}(-x) = \dots\dots\dots$
(a) $\cot^{-1}x$ (b) $\pi - \tan^{-1}x$ (c) $\pi - \cot^{-1}x$ (d) $\tan^{-1}x$
63. $\tan\left(\cos^{-1}\frac{\sqrt{3}}{2}\right) =$
(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\sqrt{3}$ (d) $\frac{2}{\sqrt{3}}$
64. $\tan^{-1}\left[\frac{2A}{1-A^2}\right]$ is equal to:
(a) $\tan^{-1}A$ (b) $\tan^{-1}\left(\frac{2}{A}\right)$ (c) $2\tan^{-1}A$ (d) $\tan^{-1}\left(\frac{A}{2}\right)$
65. $\tan^{-1}(2A) =$
(a) $\tan^{-1}\left(\frac{A}{2}\right)$ (b) $\tan^{-1}\left(\frac{2}{A}\right)$ (c) $2\tan^{-1}A$ (d) Non of these

Chapter - 14**Multiple Choice Questions***(Encircle the correct answer choice)*

1. An equation containing at least one trigonometric function is called:
(a) Trigonometric function (b) Trigonometric equation
(c) Trigonometric value (d) Periodic equation
2. If $\sin x = \frac{1}{2}$, then solution in the interval $[0, 2\pi]$ is:
(a) $\left\{\frac{\pi}{6}, \frac{5\pi}{6}\right\}$ (b) $\left\{\frac{\pi}{6}, \frac{7\pi}{6}\right\}$ (c) $\left\{\frac{\pi}{3}, \frac{4\pi}{3}\right\}$ (d) $\left\{\frac{\pi}{3}, \frac{2\pi}{3}\right\}$
3. If $\cos x = \frac{1}{2}$, then reference angle is:
(a) $\frac{\pi}{3}$ (b) $-\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $-\frac{\pi}{6}$
4. If $\sin x = -\frac{1}{2}$, then reference angle is:
(a) $\frac{\pi}{3}$ (b) $-\frac{\pi}{3}$ (c) $\frac{\pi}{6}$ (d) $-\frac{\pi}{6}$

5. General Solution of $\tan x = 1$ is:

- (a) $\{\frac{\pi}{4} + n\pi, \frac{5\pi}{4} + n\pi\}$ (b) $\{\frac{\pi}{4} + 2n\pi, \frac{5\pi}{4} + 2n\pi\}$
 (c) $\{\frac{\pi}{4} + n\pi, \frac{3\pi}{4} + n\pi\}$ (d) $\{\frac{\pi}{4} + 2n\pi, \frac{3\pi}{4} + 2n\pi\}, n \in \mathbb{Z}$

6. If $\tan 2x = -1$, then solution in the interval $[0, \pi]$ is:

- (a) $\frac{\pi}{8}$ (b) $\frac{\pi}{4}$ (c) $\frac{3\pi}{8}$ (d) $\frac{3\pi}{4}$

7. If $\sin x + \cos x = 0$, then value of $x \in [0, 2\pi]$

- (a) $\{\frac{\pi}{4}, \frac{3\pi}{4}\}$ (b) $\{\frac{\pi}{4}, \frac{7\pi}{4}\}$ (c) $\{\frac{3\pi}{4}, \frac{7\pi}{4}\}$ (d) $\{\frac{\pi}{4}, \frac{-\pi}{4}\}$

8. If $\sin 2x = \frac{\sqrt{3}}{2}$, then $x \in [0, \pi]$ is

- (a) $\{\frac{\pi}{6}, \frac{5\pi}{6}\}$ (b) $\{\frac{\pi}{6}, \frac{\pi}{12}\}$ (c) $\{\frac{\pi}{6}, \frac{5\pi}{6}\}$ (d) $\{\frac{\pi}{6}, \frac{\pi}{3}\}$

9. General solution of the equation $1 + \cos x = 0$ is:

- (a) $\{\pi + 2n\pi\}$ (b) $\{\pi + n\pi\}, n \in \mathbb{Z}$
 (c) $\{-\pi + n\pi\}$ (d) non

10. General solution of $4 \sin x - 8 = 0$ is:

- (a) $\{\pi + 2n\pi\}$ (b) $\{\pi + n\pi\}, n \in \mathbb{Z}$
 (c) $\{-\pi + n\pi\}$ (d) not possible

11. If $\sin x = \cos x$, then value of x is:

- (a) $\{\frac{\pi}{4}\}$ (b) $\{\frac{\pi}{4}, \frac{5\pi}{4}\}$ (c) $\{\frac{\pi}{4}, \frac{3\pi}{4}\}$ (d) $\{\frac{3\pi}{4}, \frac{5\pi}{4}\}$

12. If $\cot \theta = \frac{1}{\sqrt{3}}$, then value of θ in $[0, \pi]$ is:

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$ (c) $\{\frac{\pi}{3}, \frac{\pi}{6}\}$ (d) Non

13. Solution of equation $2 \sin x + \sqrt{3} = 0$ in 4th Quadrant is:

- (a) $\frac{\pi}{3}$ (b) $\frac{-\pi}{3}$ (c) $\frac{-\pi}{6}$ (d) $\frac{11\pi}{6}$

14. If $\sin x = \cos x$, then General solution is:

- (a) $\{\frac{\pi}{4} + n\pi, n \in \mathbb{Z}\}$ (b) $\{\frac{\pi}{4} + 2n\pi, n \in \mathbb{Z}\}$
 (c) $\{\frac{\pi}{4} + n\pi, \frac{5\pi}{4} + n\pi, n \in \mathbb{Z}\}$ (d) $\{\frac{\pi}{4} + 2n\pi, \frac{5\pi}{4} + 2n\pi, n \in \mathbb{Z}\}$

15. If $\cos^2 x = \frac{1}{2}$, then value of x in $[0, 2\pi]$ are:
(a) $\{\frac{2\pi}{3}, \frac{4\pi}{3}\}$ (b) $\{\frac{\pi}{3}, \frac{\pi}{6}\}$
(c) $\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{\pi}{6}\}$ (d) $\{\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}\}$
16. If $4 \sin^2 x = 3$, then value of x in $[0, \pi]$ is:
(a) $\{\frac{\pi}{3}, \frac{2\pi}{3}\}$ (b) $\{\frac{\pi}{6}, \frac{5\pi}{6}\}$ (c) $\{\frac{\pi}{3}, \frac{\pi}{6}\}$ (d) $\{\frac{2\pi}{3}, \frac{5\pi}{6}\}$
17. For the general solution, we first find the solution in the interval whose length is equal to its:
(a) range (b) domain (c) co-domain (d) period
18. All trigonometric function are function
(a) periodic (b) continues (c) injective (d) bijective
19. General solution of every trigonometric equation consists of:
(a) one solution only (b) two solutions
(c) infinity many solutions (d) No real solution
20. If $\sin 2x = \cos x$, then values of x in $[0, \pi]$ are:
(a) $\{\frac{\pi}{6}, \frac{5\pi}{6}\}$ (b) $\{\frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}\}$ (c) $\{\frac{\pi}{2}, \frac{\pi}{6}\}$ (d) $\{\frac{\pi}{2}, \frac{\pi}{3}, \frac{2\pi}{3}\}$
21. If $\sin x = 0$, then solution set is:
(a) $\{0\}$ (b) $\{\pi\}$ (c) $\{n\pi, n \in \mathbb{Z}\}$ (d) $\{2\pi, n \in \mathbb{Z}\}$
22. If $\sin = 1$, then solution set is:
(a) $\{\frac{\pi}{2}\}$ (b) $\{\frac{\pi}{2} + n\pi, \frac{3\pi}{2} + n\pi, n \in \mathbb{Z}\}$
(c) $\{\frac{\pi}{2} + 2n\pi\}$ (d) $\{n\pi, n \in \mathbb{Z}\}$
23. If $\cos x = 1$, then solution set is:
(a) $\{0\}$ (b) $\{\pi\}$ (c) $\{2n\pi, n \in \mathbb{Z}\}$ (d) $\{\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}\}$
24. If $\cos x = 0$, then solution set is:
(a) $\{\frac{\pi}{2}\}$ (b) $\{\frac{3\pi}{2}\}$ (c) $\{\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$
25. If $\tan x = 0$, then solution set is:
(a) $\{0\}$ (b) $\{\pi\}$ (c) $\{n\pi, n \in \mathbb{Z}\}$ (d) $\{2\pi, n \in \mathbb{Z}\}$
26. If $\cot x = 0$, then solution set is:
(a) $\{\frac{\pi}{2}\}$ (b) $\{\frac{3\pi}{2}\}$ (c) $\{\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$

27. If $\operatorname{cosec} x = 1$, then solution set is:
(a) $\{-\frac{\pi}{2}\}$ (b) $\{\frac{3\pi}{2}\}$ (c) $\{\frac{\pi}{2} + 2n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$
28. If $\sec x = 1$, then solution set is:
(a) $\{0\}$ (b) $\{\pi\}$ (c) $\{n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$
29. If $\sin x = -1$, then solution set is:
(a) $\{-\frac{\pi}{2}\}$ (b) $\{-\frac{3\pi}{2}\}$ (c) $\{-\frac{\pi}{2} + n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$
30. If $\cos x = -1$, then solution set is:
(a) $\{\pi\}$ (b) $\{\pi + n\pi, n \in \mathbb{Z}\}$
(c) $\{\pi + 2n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$
31. If $\operatorname{cosec} x = -1$, then solution set is:
(a) $\{-\frac{\pi}{2}\}$ (b) $\{-\frac{3\pi}{2}\}$ (c) $\{-\frac{\pi}{2} + n\pi, n \in \mathbb{Z}\}$ (d) $\{2n\pi, n \in \mathbb{Z}\}$
32. If $\sec x = -1$, then solution set is =
(a) $\{0\}$ (b) $\{\pi\}$ (c) $\{2n\pi, n \in \mathbb{Z}\}$ (d) $\{\pi + 2n\pi, n \in \mathbb{Z}\}$
33. If $\tan 4x = 1$, then value of x in $[0, 2\pi]$ is:
(a) $\{\frac{\pi}{4}, \frac{5\pi}{4}\}$ (b) $\{\frac{\pi}{16}, \frac{5\pi}{16}\}$ (c) $\{\frac{\pi}{8}, \frac{5\pi}{8}\}$ (d) $\{\frac{\pi}{8}, \frac{\pi}{16}\}$
34. One solution of $\sec x = -2$ is:
(a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $\frac{4\pi}{5}$ (d) $-\frac{\pi}{3}$
35. If $\cos \theta = -\frac{1}{2}$ and $\sin \theta = \frac{\sqrt{3}}{2}$, then θ is:
(a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $\frac{4\pi}{3}$ (d) $\frac{5\pi}{3}$
36. $\sin 2x = \frac{\sqrt{3}}{2}$ has two values of x in the interval:
(a) $[0, \frac{\pi}{2}]$ (b) $[0, 2\pi]$ (c) $[-\pi, \frac{\pi}{2}]$ (d) $[\frac{-\pi}{2}, 0]$
37. Solution of $\sin x = \frac{1}{2}$ in $[0, \pi]$ is:
(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}$ (d) $-\frac{\pi}{6}$

ANSWERS

Chapter - 1

ANSWERS KEY

(1) d (2) b (3) d (4) a (5) b (6) a (7) b (8) a (9) b (10) a
 (11) b (12) a (13) a (14) a (15) b (16) d (17) d (18) e (19) a (20) b
 (21) c (22) b (23) a (24) c (25) a (26) d (27) c (28) d (29) a (30) c
 (31) c (32) b (33) b (34) c (35) a (36) c (37) b (38) c (39) d (40) a
 (41) a (42) a (43) a (44) c (45) a (46) c (47) d (48) a (49) b (50) b
 (51) d (52) a (53) a (54) b (55) b (56) a (57) a (58) c (59) b (60) a
 (61) b (62) a (63) a (64) a (65) d (66) c (67) a (68) b (69) b (70) a
 (71) c (72) a (73) a (74) b (75) c (76) a (77) b (78) b (79) a (80) a
 (81) c (82) c (83) b (84) a

Chapter- 2

ANSWER KEY

(1) a (2) b (3) a (4) a (5) d (6) b (7) a (8) b (9) a (10) b
 (11) a (12) a (13) b (14) b (15) a (16) a (17) a (18) c (19) b (20) a
 (21) c (22) d (23) a (24) b (25) c (26) c (27) a (28) a (29) c (30) b
 (31) a (32) d (33) d (34) a (35) b (36) a (37) a (38) d (39) d (40) c
 (41) d (42) e (43) e (44) b (45) b (46) b (47) a (48) b (49) d (50) a
 (51) b (52) a (53) a (54) a (55) b (56) a (57) a (58) c (59) d (60) a
 (61) d (62) b (63) a (64) b (65) a (66) b (67) c (68) b (69) a (70) b
 (71) a (72) a (73) a (74) a (75) a (76) b (77) d (78) b (79) a (80) d
 (81) c (82) a (83) a (84) b (85) b (86) b (87) a (88) a (89) b (90) c
 (91) d (92) d (93) a (94) b (95) c (96) c (97) a (98) b (99) a (100) d
 (101) a (102) b (103) c (104) b (105) b (106) b (107) a (108) a (109) c (110) a
 (111) b (112) c (113) a (114) a (115) b (116) c (117) c (118) c (119) c (120) c
 (121) c (122) c (123) b (124) a (125) d (126) a (127) c (128) a (129) b (130) b
 (131) c (132) b (133) a (134) b (135) b (136) a (137) c (138) c (139) a (140) c, a
 (141) a (142) b

Chapter- 3

ANSWERS KEY

(1) a (2) a (3) b (4) a (5) a (6) a (7) a (8) b (9) c (10) d
 (11) b (12) a (13) a (14) d (15) a (16) b (17) a (18) d (19) a (20) d
 (21) d (22) a (23) a (24) b (25) b (26) a (27) a (28) d (29) c (30) d
 (31) b (32) b (33) d (34) b (35) a (36) d (37) a (38) a (39) b (40) b
 (41) a (42) d (43) c (44) a (45) d (46) c (47) d (48) d (49) a (50) b
 (51) d (52) a (53) d (54) d (55) d (56) b (57) a (58) b (59) c (60) d
 (61) b (62) b (63) a (64) d (65) b (66) a (67) b (68) a (69) b (70) c
 (71) d (72) c (73) a (74) b (75) b (76) a (77) a (78) d (79) b (80) a
 (81) c (82) a (83) a (84) c

Chapter- 4**ANSWERS KEY**

(1) b (2) b (3) a (4) a (5) a (6) c (7) c (8) b (9) a (10) c
 (11) c (12) a (13) c (14) b (15) c (16) b (17) c (18) c (19) a (20) c
 (21) d (22) a (23) c (24) b (25) d (26) b (27) d (28) c (29) a (30) a
 (31) b (32) d (33) c (34) a (35) a (36) d (37) c (38) d (39) c (40) a
 (41) c (42) a (43) a (44) b (45) c (46) a (47) b (48) c (49) c (50) d
 (51) b (52) b (53) a (54) b (55) a (56) c (57) d (58) c (59) d (60) b
 (61) a (62) a (63) a (64) d (65) a (66) d (67) c (68) b

Chapter- 5**ANSWERS KEY**

(1) a (2) b (3) a (4) b (5) b (6) b (7) a (8) b (9) b (10) a
 (11) a (12) d (13) c (14) b (15) b (16) b (17) c (18) d (19) b (20) a
 (21) a (22) a (23) c (24) a (25) a (26) d (27) b (28) c (29) d (30) a
 (31) b (32) a

Chapter- 6**ANSWERS KEY**

(1) a (2) b (3) c (4) a (5) c (6) d (7) a (8) d (9) b (10) a
 (11) b (12) a (13) d (14) b (15) b (16) a (17) a (18) a (19) a (20) b
 (21) b (22) b (23) a (24) a (25) c (26) a (27) a (28) b (29) b (30) b
 (31) a (32) d (33) d (34) d (35) d (36) a (37) b (38) a (39) d (40) b
 (41) a (42) a (43) a (44) b (45) d (46) d (47) a (48) b (49) a (50) a
 (51) d (52) b (53) b (54) a (55) b (56) c (57) b (58) a (59) a (60) a
 (61) a (62) a (63) c (64) a (65) d (66) a (67) b (68) b (69) c (70) c
 (71) c (72) c (73) c

Chapter- 7**ANSWERS KEY**

(1) a (2) a (3) b (4) d (5) a (6) a (7) c (8) b (9) b (10) b
 (11) a (12) b (13) a (14) a (15) a (16) a (17) d (18) c (19) b (20) d
 (21) a (22) b (23) b (24) c (25) b (26) b (27) c (28) a (29) b (30) b
 (31) a (32) b (33) c (34) a (35) b (36) a (37) a (38) a (39) a (40) a
 (41) b (42) c (43) b (44) a (45) b (46) c (47) b (48) b (49) a (50) c
 (51) a (52) b (53) b (54) b (55) b (56) b (57) a (58) a (59) c (60) b
 (61) a (62) b (63) b (64) a (65) a (66) a (67) a (68) b (69) c (70) a
 (71) c (72) d (73) a (74) d (75) c (76) d (77) a

Chapter - 8**ANSWERS KEY**

(1) c (2) d (3) b (4) b (5) a (6) b (7) a (8) b (9) c (10) b
 (11) a (12) c (13) a (14) b (15) a (16) b (17) b (18) b (19) b (20) c
 (21) b (22) d (23) b (24) b (25) a (26) b (27) d (28) d (29) b (30) a
 (31) d (32) a (33) b (34) d (35) c

Chapter- 9**ANSWERS KEY**

(1) b (2) c (3) b (4) d (5) c (6) d (7) d (8) b (9) b (10) b
 (11) b (12) b (13) b (14) a (15) c (16) c (17) b (18) c (19) c (20) d
 (21) b (22) c (23) a (24) a (25) a (26) b (27) a (28) b (29) a (30) a
 (31) a (32) c (33) b (34) a (35) c (36) a (37) b (38) b (39) b (40) c
 (41) b (42) d (43) a (44) b (45) c (46) a (47) b (48) c (49) d (50) b
 (51) a (52) a (53) c (54) b (55) d (56) b (57) a (58) a (59) a (60) b
 (61) d (62) b (63) d

Chapter- 10**ANSWERS KEY**

(1) a (2) a (3) b (4) a (5) b (6) d (7) c (8) c (9) d (10) b
 (11) a (12) d (13) d (14) b (15) c (16) a (17) a (18) b (19) d (20) a
 (21) c (22) a (23) c (24) d (25) d (26) c (27) a (28) d (29) d (30) c
 (31) d (32) d (33) b (34) c (35) a (36) b (37) c (38) a (39) c (40) b
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 (51) c (52) b (53) a (54) c (55) a (56) a (57) b (58) c (59) d (60) a
 (61) a (62) a (63) b (64) a (65) b (66) b (67) c (68) a

Chapter- 11**ANSWERS KEY**

(1) a (2) a (3) c (4) b (5) c (6) b (7) b (8) b (9) a (10) a
 (11) b (12) b (13) d (14) b (15) b (16) a (17) a (18) b (19) b (20) c
 (21) b (22) a (23) b (24) a (25) c (26) c (27) a (28) a (29) c (30) d
 (31) b (32) d (33) b (34) d (35) d (35) b

Chapter- 12**ANSWERS KEY**

(1) d (2) b (3) a (4) b (5) a (6) d (7) a (8) b (9) c (10) c
 (11) a (12) c (13) d (14) c (15) b (16) d (17) b (18) d (19) d (20) d
 (21) c (22) a (23) d (24) d (25) b (26) b (27) d (28) c (29) d (30) c
 (31) c (32) a (33) a (34) a (35) c (36) c (37) b (38) b (39) c (40) a
 (41) a (42) b (43) b (44) d (45) d (46) c (47) c (48) a (49) d (50) c
 (51) a (52) d (53) a (54) c (55) a

Chapter- 13**ANSWERS KEY**

(1) a (2) b (3) c (4) d (5) b (6) c (7) a (8) a (9) b (10) b
(11) c (12) c (13) a (14) a (15) b (16) b (17) c (18) c (19) a (20) b
(21) c (22) d (23) a (24) b (25) b (26) a (27) a (28) b (29) a (30) a
(31) b (32) a (33) b (34) b (35) a (36) a (37) d (38) a (39) d (40) c
(41) a (42) d (43) a (44) b (45) d (46) a (47) d (48) a (49) b (50) a
(51) b (52) a (53) b (54) b (55) a (56) c (57) a (58) a (59) a (60) a
(61) c (62) c (63) a (64) c (65) d

Chapter- 14**ANSWERS KEY**

(1) b (2) a (3) a (4) c (5) a (6) a (7) c (8) d (9) a (10) d
(11) b (12) a (13) b (14) c (15) d (16) a (17) d (18) a (19) c (20) b
(21) c (22) c (23) c (24) c (25) c (26) c (27) c (28) d (29) a (30) c
(31) a (32) d (33) b (34) a (35) c (36) a (37) c